Software Requirements Specification

for

Blockchain-Based Certificate Verification System

Version 1.0 approved

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Organization: IEEE

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for the **Blockchain-Based Certificate Verification System**. This system is designed to provide a secure, immutable, and publicly verifiable method for issuing and verifying academic certificates using blockchain technology. The system aims to address the growing problem of certificate fraud and counterfeiting by leveraging the transparency and immutability of blockchain. It is intended to be used by educational institutions, employers, and students to streamline the certificate verification process, reduce manual intervention, and ensure the authenticity of academic credentials.

1.2 Document Conventions

This document follows the IEEE SRS template. Requirements are numbered sequentially and are categorized by system features. High-priority requirements are marked with an asterisk (*). The document uses the following typographical conventions:

- **Bold text** for section headings and key terms.
- *Italic text* for emphasis or special notes.
- Monospace font for code snippets or technical terms.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- **Developers:** Responsible for implementing the system based on the requirements outlined in this document.
- **Project Managers:** Oversee the development and deployment of the system.
- **Educational Institutions:** Will use the system to issue and verify academic certificates.

- **Employers:** Will use the system to verify the academic credentials of job applicants.
- **Students:** Will use the system to share their academic certificates with potential employers.

It is recommended that readers start with the **Introduction** and **Overall Description** sections to gain a high-level understanding of the system before diving into the detailed system features and requirements.

1.4 Product Scope

The **Blockchain-Based Certificate Verification System** is a cloud-based solution that leverages blockchain technology to issue and verify academic certificates. The system is designed to replace traditional manual verification processes, which are time-consuming and prone to errors. The system will be used by educational institutions to issue digital certificates, by employers to verify the authenticity of these certificates, and by students to share their academic credentials with potential employers.

The system will provide the following key benefits:

- **Immutability:** Certificates stored on the blockchain cannot be altered or tampered with, ensuring their authenticity.
- **Transparency:** All transactions on the blockchain are publicly verifiable, providing a high level of trust in the system.
- **Efficiency:** The system automates the verification process, reducing the time and effort required to verify certificates.
- **Security:** The use of cryptographic hashing and smart contracts ensures that the system is secure and resistant to fraud.

1.5 References

- IEEE SRS Template
- Blockchain Technology in Education: A Review by D. Ji et al.
- Smart Contracts and Blockchain by Curmi et al.
- Blockchain-Based Cybersecurity by Dasgupta et al.
- SHA-256 Hashing Algorithm Documentation
- Solidity Programming Language Documentation

2. Overall Description

2.1 Product Perspective

The **Blockchain-Based Certificate Verification System** is a new, self-contained product designed to replace traditional manual certificate verification processes. It integrates with existing educational systems and provides a secure, decentralized method for issuing and verifying academic certificates. The system is built on blockchain technology, which ensures that all certificates are immutable and publicly verifiable. The system is designed to be used by educational institutions, employers, and students, and it will be accessible via a web-based interface.

2.2 Product Functions

The system will provide the following key functions:

- **Certificate Issuance:** Educational institutions can issue digital certificates to students upon course completion. The certificates will be stored on the blockchain, ensuring their immutability and authenticity.
- **Certificate Verification:** Employers and institutions can verify the authenticity of certificates using blockchain technology. The system will provide a quick and efficient method for verifying certificates, reducing the time and effort required for manual verification.
- **Smart Contract Execution:** The system will use smart contracts to automate the verification process. Smart contracts will be executed on the blockchain, ensuring that the verification process is secure and tamper-proof.
- **QR Code Generation:** The system will generate QR codes for each certificate, allowing for quick and secure verification. The QR codes will contain encoded information about the certificate, which can be scanned and verified using a smartphone or QR code scanner.

2.3 User Classes and Characteristics

The system will be used by the following user classes:

- **Students:** Users who receive and share their academic certificates. Students will use the system to upload their certificates and share them with potential employers.
- **Educational Institutions:** Users who issue certificates and verify their authenticity. Educational institutions will use the system to issue digital certificates to students and verify the authenticity of certificates issued by other institutions.
- **Employers:** Users who verify the academic credentials of job applicants. Employers will use the system to verify the authenticity of certificates submitted by job applicants.

2.4 Operating Environment

The system will operate in a cloud-based environment and will be compatible with major operating systems (Windows, macOS, Linux). It will require internet access and will support modern web browsers. The system will be built using the following technologies:

- **Blockchain:** Ethereum or Hyperledger for certificate storage and verification.
- **Smart Contracts:** Written in Solidity for automated verification.
- **Web Development:** Next.js for the front-end interface.
- **Database:** Firebase for real-time data management.
- **Backend:** Node.js SDK for server-side logic.

2.5 Design and Implementation Constraints

The system must adhere to the following design and implementation constraints:

- **Blockchain Technology:** The system must use blockchain technology for immutability and transparency.
- **Smart Contracts:** Smart contracts must be written in Solidity and deployed on a blockchain network.
- **QR Code Generation:** The system must support QR code generation for certificate verification.
- **Cryptographic Hashing:** The system must use SHA-256 hashing to ensure data integrity.
- **Scalability:** The system must be scalable to support a large number of users and certificates.

2.6 User Documentation

The following user documentation will be provided:

- **User Manual:** A comprehensive guide for using the system, including instructions for issuing and verifying certificates.
- **Online Help:** Context-sensitive help available within the system.
- **Tutorials:** Step-by-step tutorials for common tasks, such as issuing certificates and verifying certificates.
- **API Documentation:** Documentation for developers who wish to integrate the system with other applications.

2.7 Assumptions and Dependencies

The system makes the following assumptions:

- **Accurate Data:** Educational institutions will provide accurate certificate data.
- **Blockchain Network:** The system depends on a blockchain network for secure and immutable record-keeping.
- **Internet Access:** Users will have reliable internet access to use the system.

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3. External Interface Requirements

3.1 User Interfaces

The system will provide a web-based interface for users to issue and verify certificates. The interface will include the following components:

- **Dashboard:** A dashboard for educational institutions to issue certificates and view issued certificates.
- **Verification Portal:** A portal for employers to verify the authenticity of certificates.
- **Student Portal:** A portal for students to upload and share their certificates with potential employers.

3.2 Hardware Interfaces

The system will interface with standard hardware devices such as scanners and printers for QR code generation and verification. The system will also support the use of smartphones for scanning QR codes.

3.3 Software Interfaces

The system will integrate with the following software components:

- **Blockchain Networks:** Ethereum or Hyperledger for certificate storage and verification.
- **Cloud Storage:** Firebase for real-time data management.
- **Web Browsers:** The system will be compatible with modern web browsers such as Chrome, Firefox, and Safari.

3.4 Communications Interfaces

The system will use HTTPS for secure communication between clients and servers. It will also support RESTful APIs for integration with third-party systems.

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4. System Features

4.1 Certificate Issuance and Verification

4.1.1 Description and Priority

This feature allows educational institutions to issue digital certificates and employers to verify them. It is of high priority.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** Student completes a course.
- **Response:** Institution issues a digital certificate and stores it on the blockchain.

4.1.3 Functional Requirements

- **REQ-1:** The system must allow institutions to issue digital certificates.
- **REQ-2:** The system must allow employers to verify certificates using blockchain technology.

4.2 Smart Contract Execution

4.2.1 Description and Priority

This feature automates the verification process using smart contracts. It is of high priority.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Employer requests certificate verification.
- **Response:** Smart contract executes and returns verification results.

4.2.3 Functional Requirements

- **REQ-3:** The system must execute smart contracts for certificate verification.
- **REQ-4:** The system must generate QR codes for verified certificates.

4.3 QR Code Generation and Verification

4.3.1 Description and Priority

This feature generates QR codes for certificates and allows for quick verification. It is of medium priority.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** Employer scans a QR code.
- **Response:** System displays certificate details and verification status.

4.3.3 Functional Requirements

- **REQ-5:** The system must generate QR codes for each certificate.
- **REQ-6:** The system must allow users to scan QR codes for verification.

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** The system must process certificate verification requests within **5 seconds** under normal operating conditions. This ensures that employers and institutions can quickly verify certificates without significant delays.
- **Throughput:** The system must support at least **100 concurrent verification requests** without degradation in performance.
- **Database Query Performance:** Database queries for retrieving certificate information from the blockchain must execute within **2 seconds**.

5.2 Safety Requirements

- **Data Encryption:** All sensitive data, including certificate information and user credentials, must be encrypted during transmission and storage using **AES-256 encryption**.
- **Data Integrity:** The system must use **cryptographic hashing (e.g., SHA-256)** to ensure the integrity of certificate data. Any tampering with the data will result in a mismatch between the stored hash and the computed hash.
- **Backup and Recovery:** The system must implement regular backup procedures to ensure that data can be recovered in the event of a failure or data loss.

5.3 Security Requirements

- **Authentication:** The system must implement **multi-factor authentication (MFA)** for all users, including students, educational institutions, and employers.
- **Authorization:** The system must enforce role-based access control (RBAC) to ensure that only authorized users can perform specific actions.
- **Audit Logging:** The system must maintain detailed audit logs of all user actions, including certificate issuance, verification, and modifications.

5.4 Software Quality Attributes

Availability: The system must have a **99.9% uptime**, ensuring that it is accessible to users at all times.

Scalability: The system must be scalable to support a **large number of users** (e.g., thousands of concurrent users) and certificates without performance degradation.

Usability: The system must provide an intuitive and user-friendly interface, with clear instructions and minimal learning curve for users.

Maintainability: The system must be designed for ease of maintenance, with modular code and comprehensive documentation.

5.5 Business Rules

Certificate Issuance: Only **authorized educational institutions** can issue certificates. Institutions must be registered and verified by the system administrator before they can issue certificates.

Certificate Verification: Only **authorized employers** can verify certificates. Employers must be registered and verified by the system administrator before they can access the verification feature.

Data Retention: Certificate data must be retained for a minimum of **10 years** to comply with legal and regulatory requirements.

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6. Other Requirements

- The system must comply with international data protection regulations (e.g., GDPR).

Appendix A: Glossary

- **Blockchain:** A decentralized digital ledger used to record transactions.
- **Smart Contract:** Self-executing contracts with the terms of the agreement directly written into code.
- **QR Code:** A machine-readable code consisting of black and white squares, used for storing information.

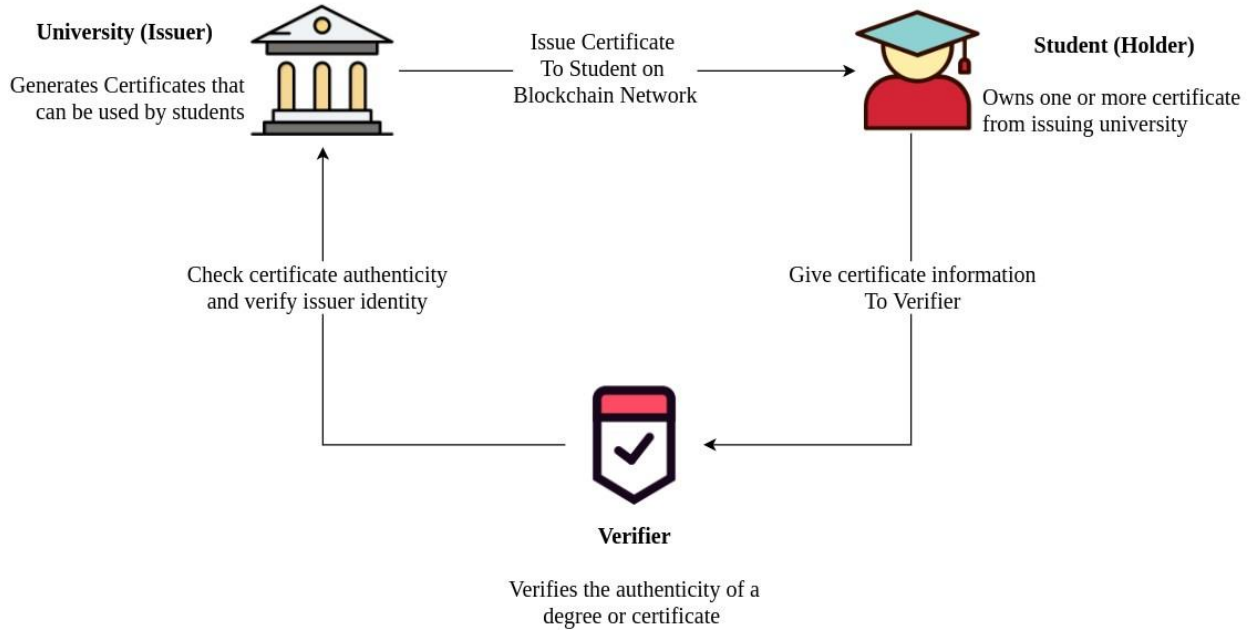
Appendix B: Analysis Models

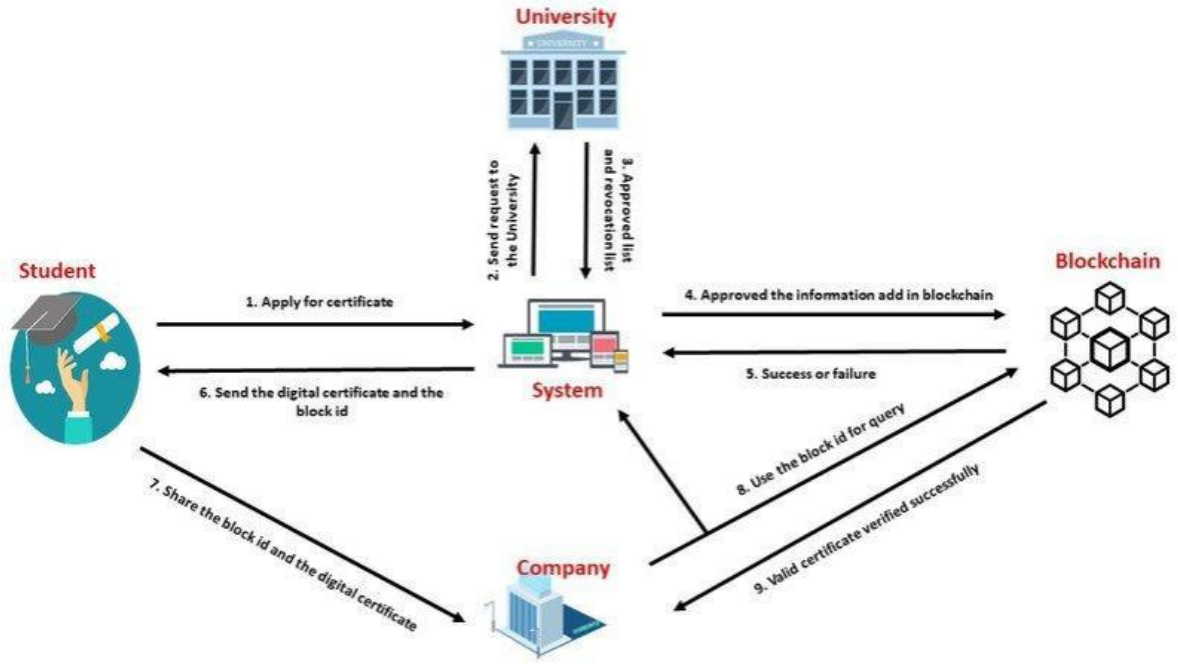
- Data Flow Diagram
- Use Case Diagram

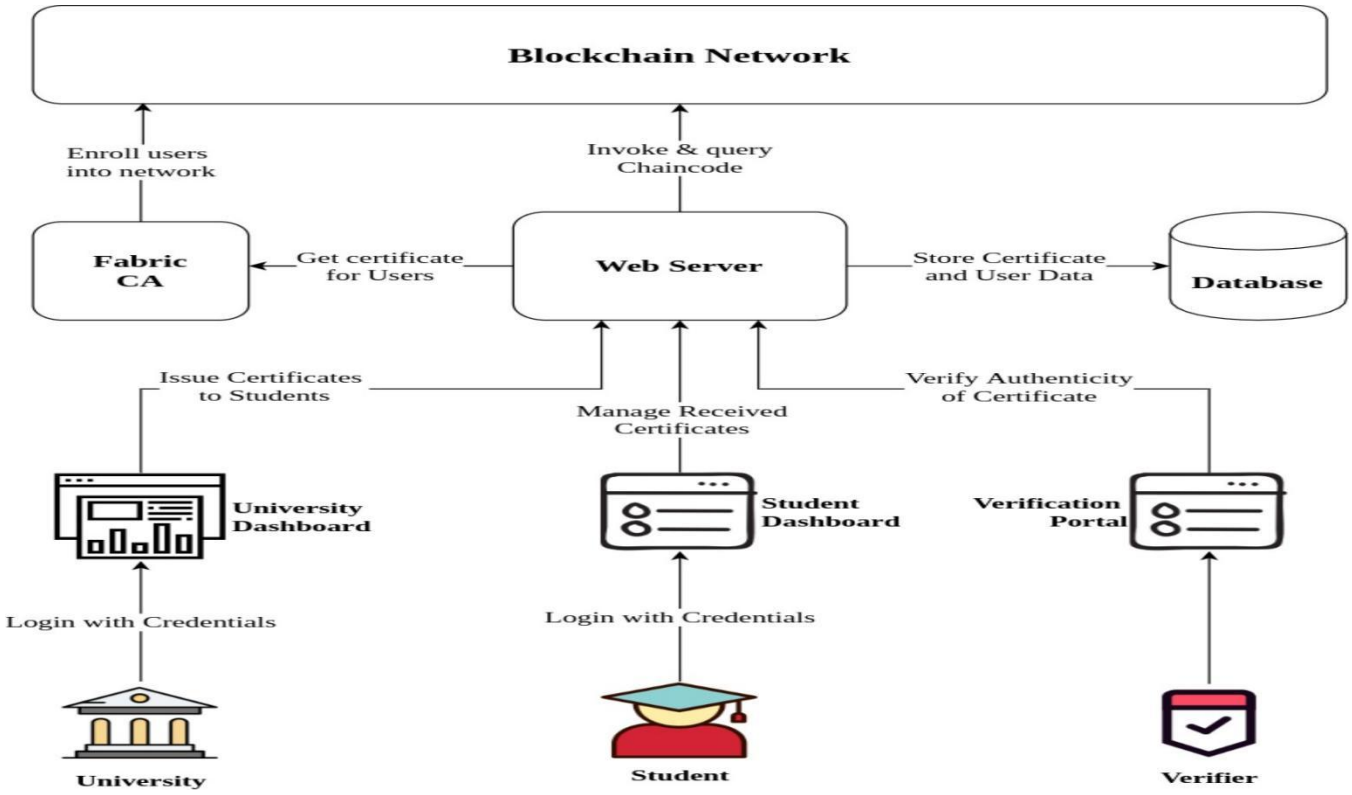
Appendix C: To Be Determined List

- TBD-1: Finalize the blockchain network to be used (Ethereum, Hyperledger, etc.).
- TBD-2: Determine the exact cryptographic hashing algorithm to be used.

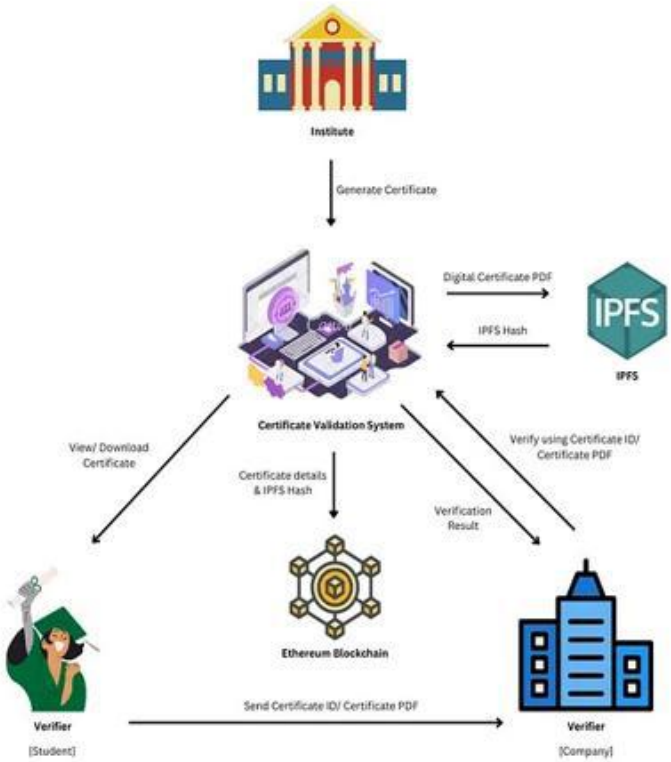
Flowchart of the system







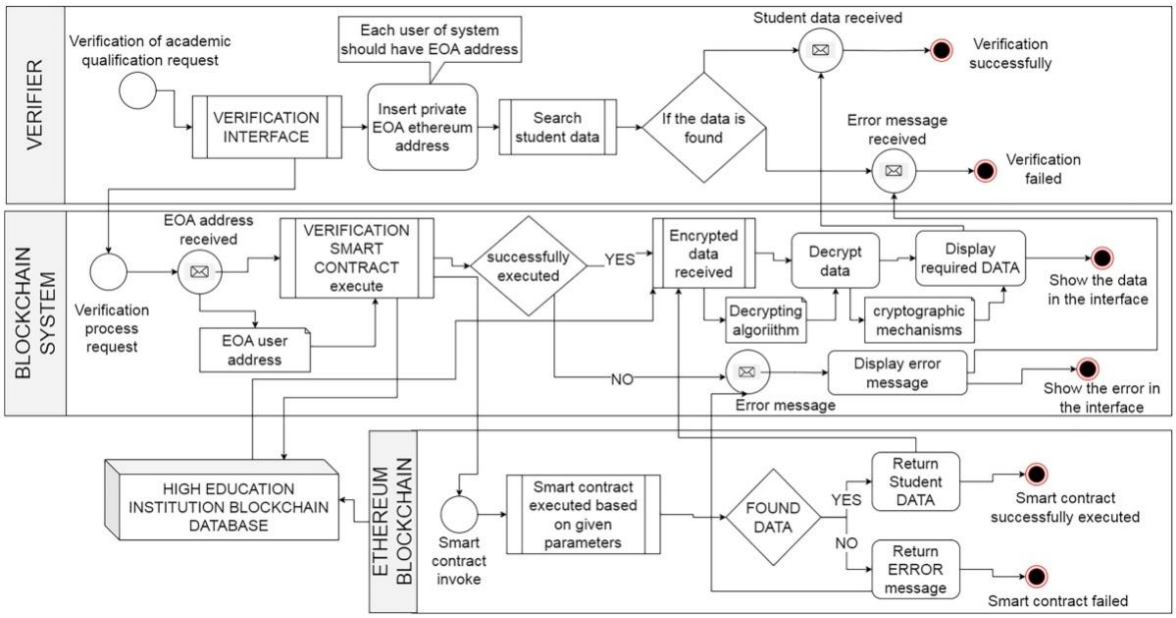
Use Case Diagrams



Design - UML Diagrams

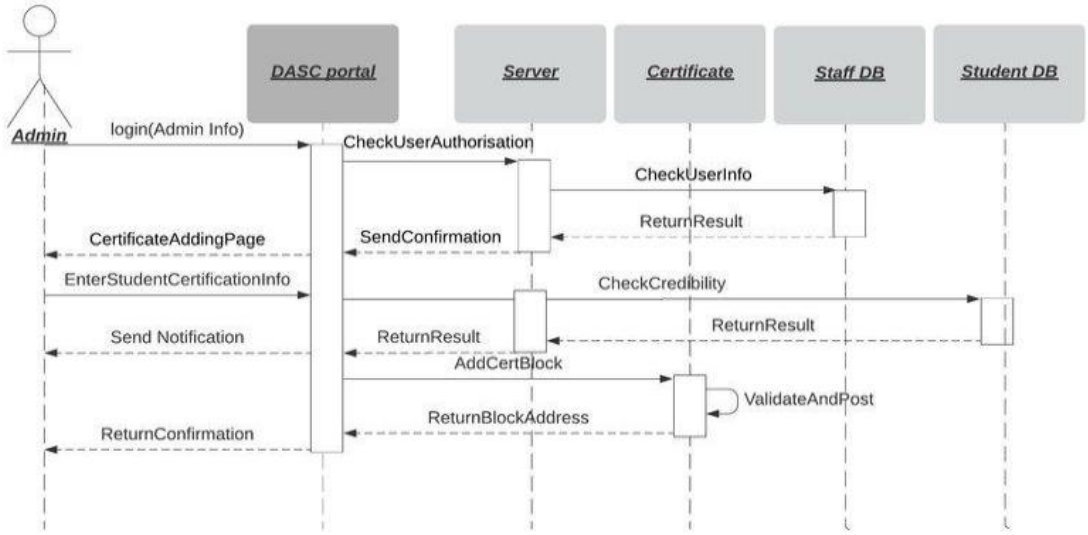
Class Diagram:

Represents the structure of the system, including classes like Certificate, NFT, User, and Blockchain.



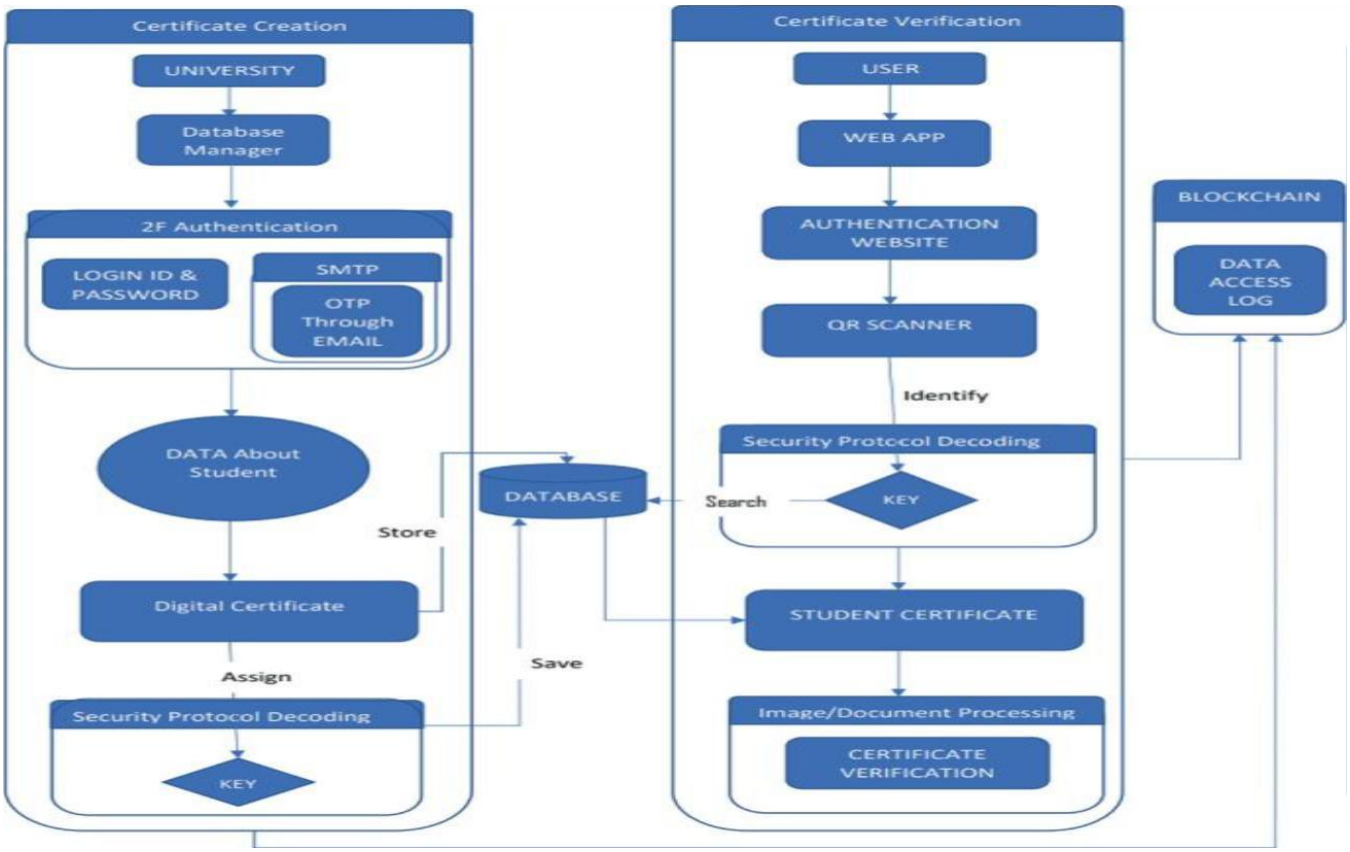
Sequence Diagram:

Illustrates the step-by-step flow from certificate issuance to verification.



Activity Diagram:

Shows the workflow of user interactions and system processes.



Conclusion:

Project Summary:

- A secure, blockchain-based system for academic credential verification.
- Combines blockchain immutability, AI-powered fraud detection, and NFT-based ownership tracking.
- Ensures trust, transparency, and efficiency in credential verification.

Impact:

- Reduces fraud, saves time, and enhances trust in academic credentials.

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   3.4 Communications Interfaces

4. **System Features**
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   4.3 QR Code Generation and Verification

5. **Other Nonfunctional Requirements**
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5.4 Software Quality Attributes
5.5 Business Rules

6. **Other Requirements**
Appendix A: Glossary
Appendix B: Analysis Models
Appendix C: To Be Determined List

1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for the Blockchain-Based Certificate Verification System. This system aims to provide a secure, immutable, and publicly verifiable method for issuing and verifying academic certificates using blockchain technology. The system is designed to reduce fraud and counterfeiting in digital certificates, ensuring quick and anti-counterfeit verification.

1.2 Document Conventions

This document follows the IEEE SRS template. Requirements are numbered sequentially and are categorized by system features. High-priority requirements are marked with an asterisk (*). The document uses the following typographical conventions:

- Bold text for section headings and key terms.
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- Monospace font for code snippets or technical terms.

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This document is intended for the following audiences:

- Developers: Responsible for implementing the system based on the requirements outlined in this document.
- Project Managers: Oversee the development and deployment of the system.
- Educational Institutions: Will use the system to issue and verify academic certificates.
- Employers: Will use the system to verify the academic credentials of job applicants.
- Students: Will use the system to share their academic certificates with potential employers.

It is recommended that readers start with the Introduction and Overall Description sections to gain a high-level understanding of the system before diving into the detailed system features and requirements.

1.4 Product Scope

The Blockchain-Based Certificate Verification System is a cloud-based solution that leverages blockchain technology to issue and verify academic certificates. The system is designed to replace traditional manual verification processes, which are time-consuming

and prone to errors. The system will be used by educational institutions to issue digital certificates, by employers to verify the authenticity of these certificates, and by students to share their academic credentials with potential employers.
The system will provide the following key benefits:

- Immutability: Certificates stored on the blockchain cannot be altered or tampered with, ensuring their authenticity.
- Transparency: All transactions on the blockchain are publicly verifiable, providing a high level of trust in the system.
- Efficiency: The system automates the verification process, reducing the time and effort required to verify certificates.
- Security: The use of cryptographic hashing and smart contracts ensures that the system is secure and resistant to fraud.

1.5 References

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- Smart Contracts and Blockchain by Curmi et al.
- Blockchain-Based Cybersecurity by Dasgupta et al.
- SHA-256 Hashing Algorithm Documentation
- Solidity Programming Language Documentation

2. Overall Description

2.1 Product Perspective

The Blockchain-Based Certificate Verification System is a new, self-contained product designed to replace traditional manual certificate verification processes. It integrates with existing educational systems and provides a secure, decentralized method for issuing and verifying academic certificates. The system is built on blockchain technology, which ensures that all certificates are immutable and publicly verifiable. The system is designed to be used by educational institutions, employers, and students, and it will be accessible via a web-based interface.

2.2 Product Functions

The system will provide the following key functions:

- Certificate Issuance: Educational institutions can issue digital certificates to students upon course completion. The certificates will be stored on the blockchain, ensuring their immutability and authenticity.
- Certificate Verification: Employers and institutions can verify the authenticity of certificates using blockchain technology. The system will provide a quick and efficient method for verifying certificates, reducing the time and effort required for manual verification.
- Smart Contract Execution: The system will use smart contracts to automate the verification process. Smart contracts will be executed on the blockchain, ensuring

that the verification process is secure and tamper-proof.

- QR Code Generation: The system will generate QR codes for each certificate, allowing for quick and secure verification. The QR codes will contain encoded information about the certificate, which can be scanned and verified using a smartphone or QR code scanner.

2.3 User Classes and Characteristics

The system will be used by the following user classes:

- Students: Users who receive and share their academic certificates. Students will use the system to upload their certificates and share them with potential employers.
- Educational Institutions: Users who issue certificates and verify their authenticity. Educational institutions will use the system to issue digital certificates to students and verify the authenticity of certificates issued by other institutions.
- Employers: Users who verify the academic credentials of job applicants. Employers will use the system to verify the authenticity of certificates submitted by job applicants.

2.4 Operating Environment

The system will operate in a cloud-based environment and will be compatible with major operating systems (Windows, macOS, Linux). It will require internet access and will support modern web browsers. The system will be built using the following technologies:

- Blockchain: Ethereum or Hyperledger for certificate storage and verification.
- Smart Contracts: Written in Solidity for automated verification.
- Web Development: Next.js for the front-end interface.
- Database: Firebase for real-time data management.
- Backend: Node.js SDK for server-side logic.

2.5 Design and Implementation Constraints

The system must adhere to the following design and implementation constraints:

- Blockchain Technology: The system must use blockchain technology for immutability and transparency.
- Smart Contracts: Smart contracts must be written in Solidity and deployed on a blockchain network.
- QR Code Generation: The system must support QR code generation for certificate verification.
- Cryptographic Hashing: The system must use SHA-256 hashing to ensure data integrity.
- Scalability: The system must be scalable to support a large number of users and certificates.

2.6 User Documentation

The following user documentation will be provided:

- User Manual: A comprehensive guide for using the system, including instructions for issuing and verifying certificates.
- Online Help: Context-sensitive help available within the system.
- Tutorials: Step-by-step tutorials for common tasks, such as issuing certificates and verifying certificates.
- API Documentation: Documentation for developers who wish to integrate the system with other applications.

2.7 Assumptions and Dependencies

The system makes the following assumptions:

- Accurate Data: Educational institutions will provide accurate certificate data.
- Blockchain Network: The system depends on a blockchain network for secure and immutable record-keeping.
- Internet Access: Users will have reliable internet access to use the system.

3. External Interface Requirements

3.1 User Interfaces

The system will provide a web-based interface for users to issue and verify certificates. The interface will include the following components:

- Dashboard: A dashboard for educational institutions to issue certificates and view issued certificates.
- Verification Portal: A portal for employers to verify the authenticity of certificates.
- Student Portal: A portal for students to upload and share their certificates with potential employers.

3.2 Hardware Interfaces

The system will interface with standard hardware devices such as scanners and printers for QR code generation and verification. The system will also support the use of smartphones for scanning QR codes.

3.3 Software Interfaces

The system will integrate with the following software components:

- Blockchain Networks: Ethereum or Hyperledger for certificate storage and verification.
- Cloud Storage: Firebase for real-time data management.
- Web Browsers: The system will be compatible with modern web browsers such as Chrome, Firefox, and Safari.

3.4 Communications Interfaces

The system will use HTTPS for secure communication between clients and servers. It will also support RESTful APIs for integration with third-party systems.

4. System Features

4.1 Certificate Issuance and Verification

4.1.1 Description and Priority

This feature allows educational institutions to issue digital certificates and employers to verify them. It is of high priority.

4.1.2 Stimulus/Response Sequences

- Stimulus: Student completes a course.
- Response: Institution issues a digital certificate and stores it on the blockchain.

4.1.3 Functional Requirements

- REQ-1: The system must allow institutions to issue digital certificates.
- REQ-2: The system must allow employers to verify certificates using blockchain technology.

4.2 Smart Contract Execution

4.2.1 Description and Priority

This feature automates the verification process using smart contracts. It is of high priority.

4.2.2 Stimulus/Response Sequences

- Stimulus: Employer requests certificate verification.
- Response: Smart contract executes and returns verification results.

4.2.3 Functional Requirements

- REQ-3: The system must execute smart contracts for certificate verification.
- REQ-4: The system must generate QR codes for verified certificates.

4.3 QR Code Generation and Verification

4.3.1 Description and Priority

This feature generates QR codes for certificates and allows for quick verification. It is of medium priority.

4.3.2 Stimulus/Response Sequences

- Stimulus: Employer scans a QR code.
- Response: System displays certificate details and verification status.

4.3.3 Functional Requirements

- REQ-5: The system must generate QR codes for each certificate.
- REQ-6: The system must allow users to scan QR codes for verification.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system must process certificate verification requests within 5 seconds under normal operating conditions.
- The system must support at least 100 concurrent verification requests without degradation in performance.
- Database queries for retrieving certificate information from the blockchain must execute within 2 seconds.

5.2 Safety Requirements

- The system must ensure that sensitive data is encrypted during transmission and storage using AES-256 encryption.
- The system must use cryptographic hashing (e.g., SHA-256) to ensure the integrity of certificate data.
- The system must implement regular backup procedures to ensure data recovery in the event of a failure or data loss.

5.3 Security Requirements

- The system must implement multi-factor authentication (MFA) for all users.
- The system must enforce role-based access control (RBAC) to ensure that only authorized users can perform specific actions.
- The system must maintain detailed audit logs of all user actions, including certificate issuance, verification, and modifications.
- The system must undergo regular vulnerability scans to identify and address potential security weaknesses.

5.4 Software Quality Attributes

- The system must have a 99.9% uptime, ensuring high availability.
- The system must be scalable to support a large number of users and certificates.
- The system must provide an intuitive and user-friendly interface, with clear instructions and minimal learning curve for users.
- The system must be designed for ease of maintenance, with modular code and comprehensive documentation.

5.5 Business Rules

- Only authorized educational institutions can issue certificates.
- Only authorized employers can verify certificates.
- Certificate data must be retained for a minimum of 10 years to comply with legal and regulatory requirements.
- The system must comply with all relevant data protection regulations, including GDPR and FERPA.

6. Other Requirements

- The system must comply with international data protection regulations (e.g., GDPR).

Appendix A: Glossary

- Blockchain: A decentralized digital ledger used to record transactions.
- Smart Contract: Self-executing contracts with the terms of the agreement directly written into code.
- QR Code: A machine-readable code consisting of black and white squares, used for storing information.

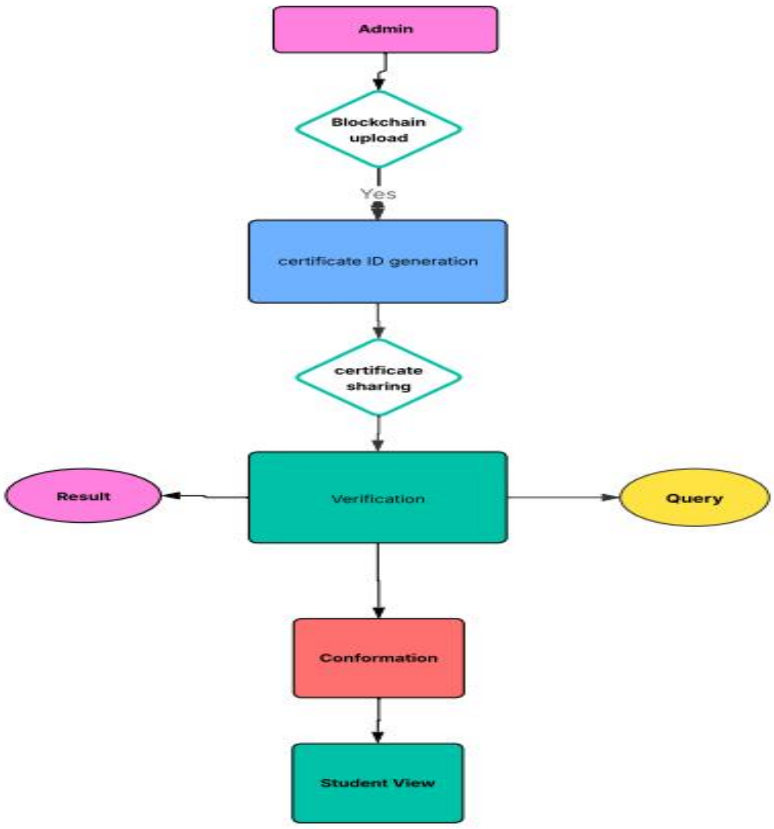
Appendix B: Analysis Models

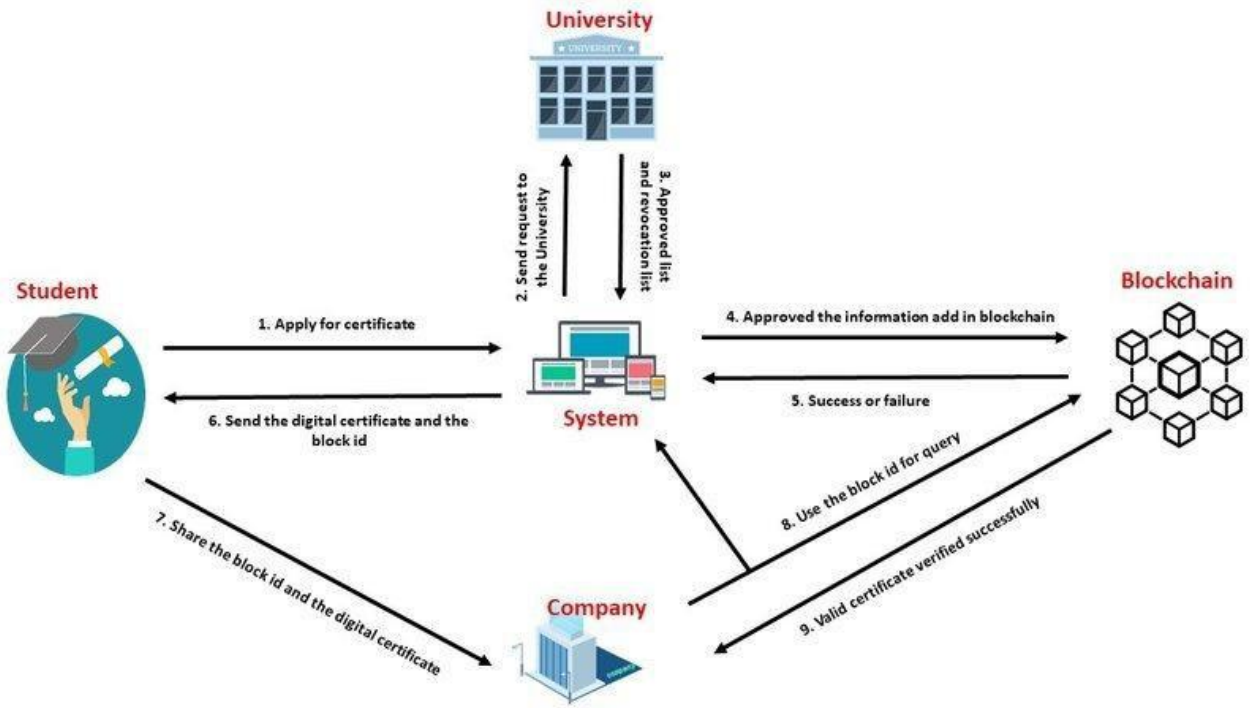
- Data Flow Diagram
- Use Case Diagram

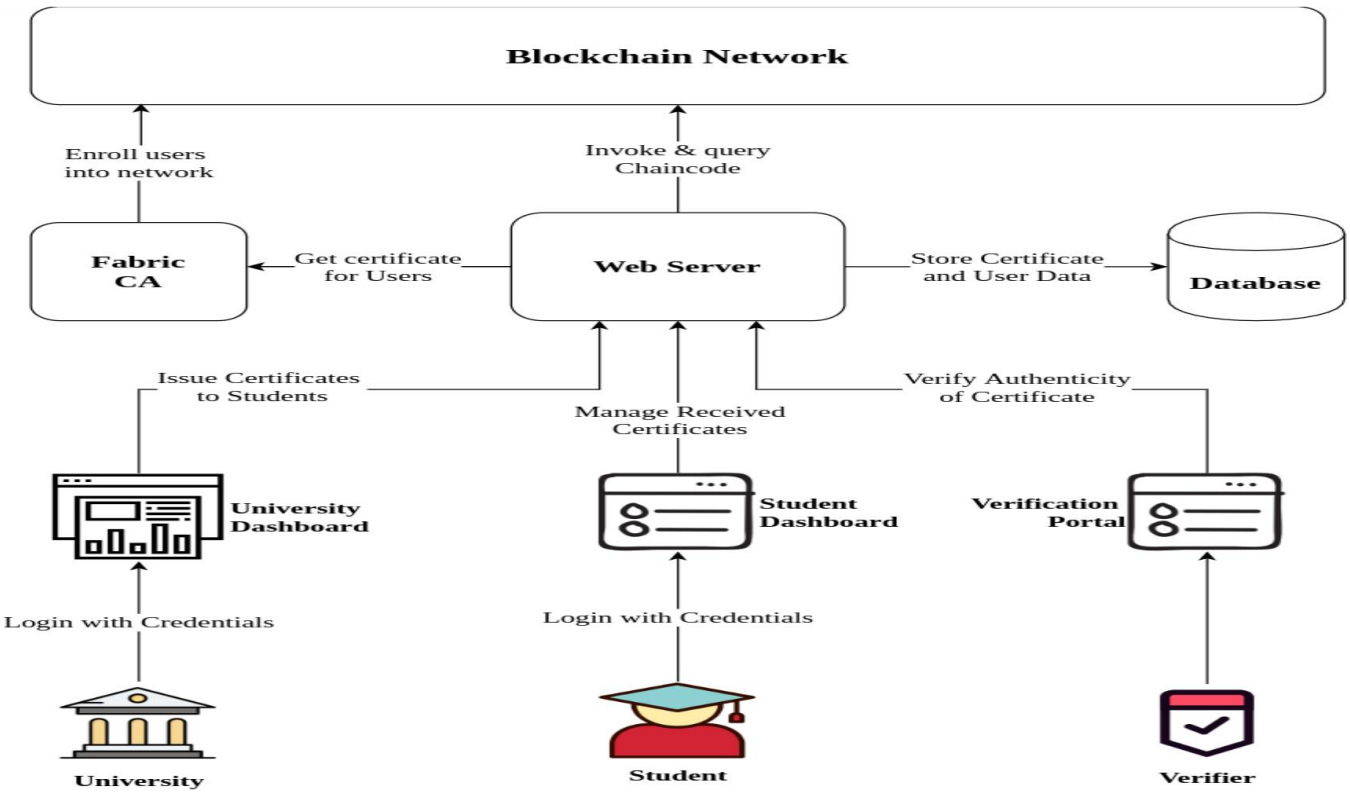
Appendix C: To Be Determined List

- TBD-1: Finalize the blockchain network to be used (Ethereum, Hyperledger, etc.).
- TBD-2: Determine the exact cryptographic hashing algorithm to be used.

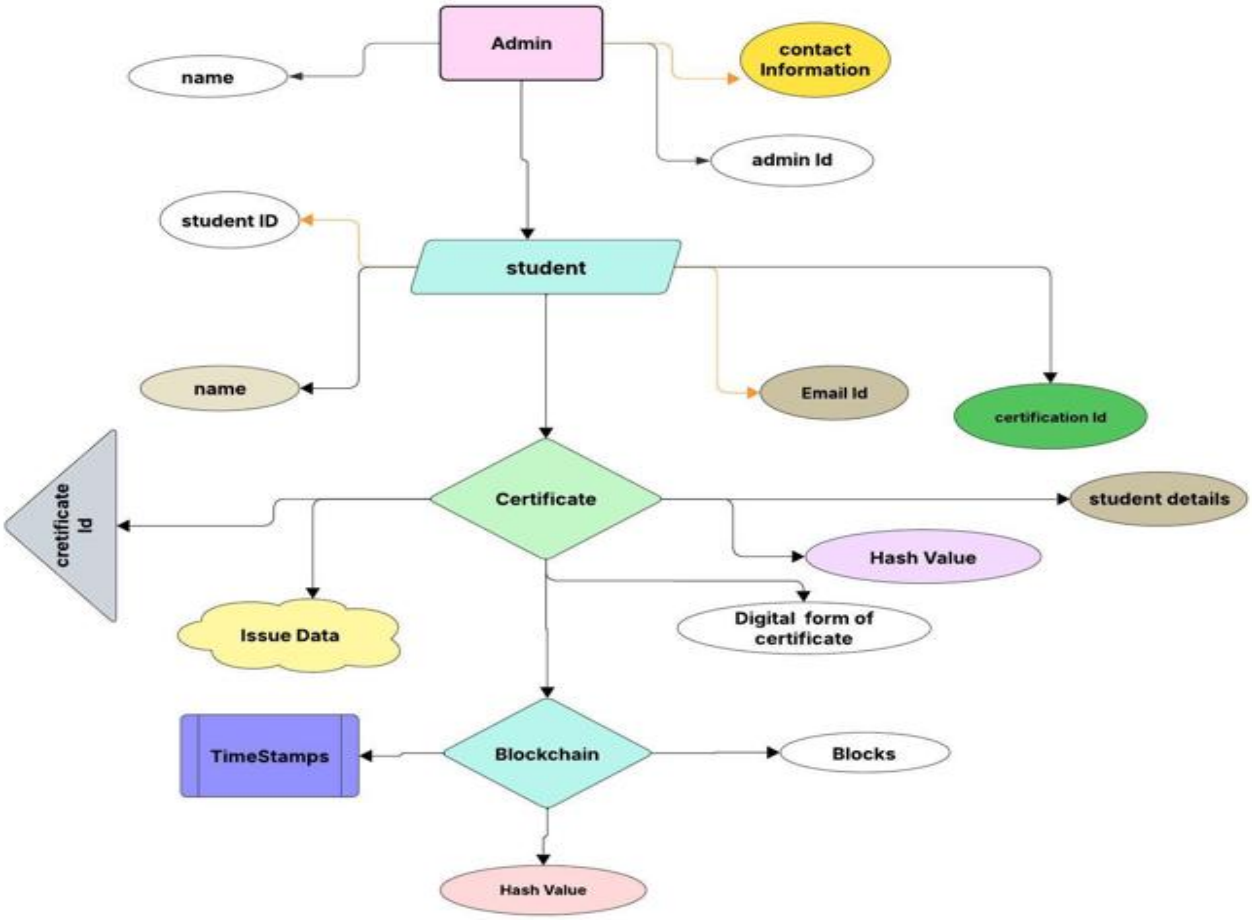
Flowchart of the system




University
2. Send request to the University
3. Approved list and revocation list
Student
1. Apply for certificate
6. Send the digital certificate and the block id
System
4. Approved the information add in blockchain
5. Success or failure
Blockchain
8. Use the block id for query
9. Valid certificate verified successfully
7. Share the block id and the digital certificate
Company


Blockchain Network
Enroll users into network
Invoke & query Chaincode
Fabric CA
Get certificate for Users
Web Server
Store Certificate and User Data
Database
Issue Certificates to Students
Manage Received Certificates
Verify Authenticity of Certificate
University Dashboard
Student Dashboard
Verification Portal
Login with Credentials
Login with Credentials
University
Student
Verifier

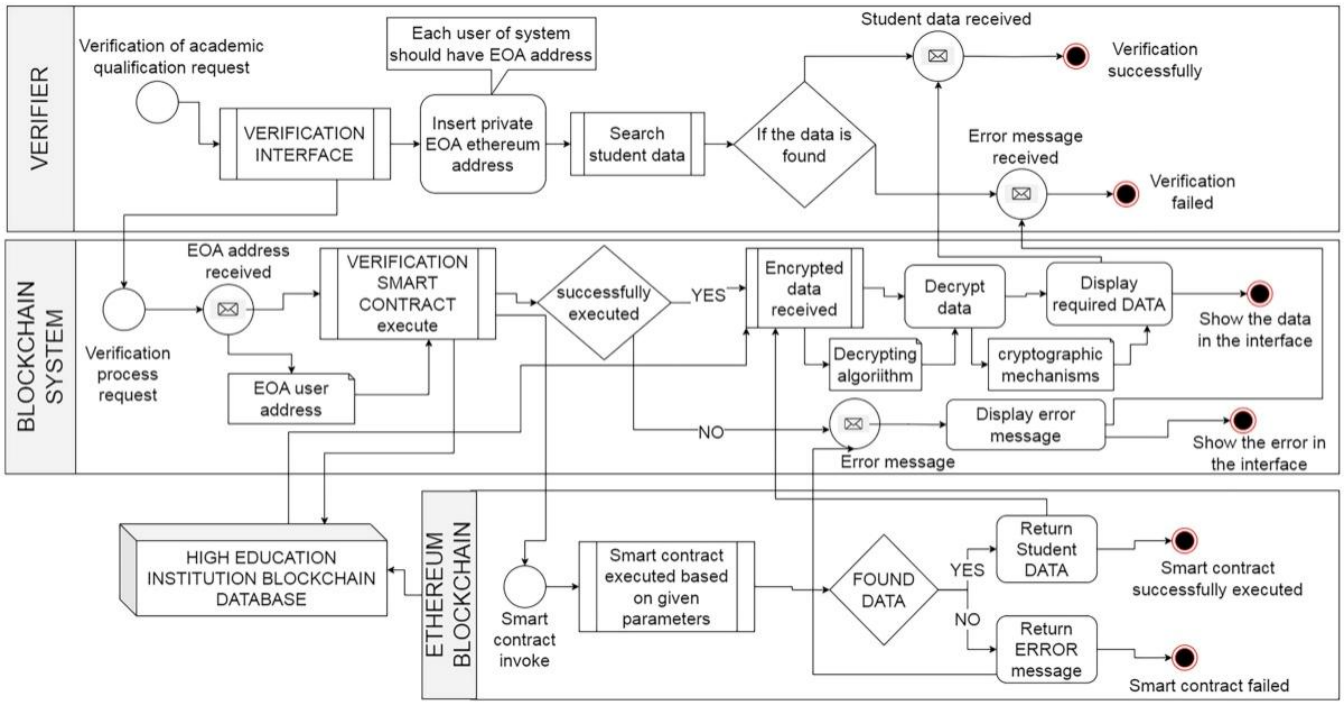
Use Case Diagrams



Design - UML Diagrams

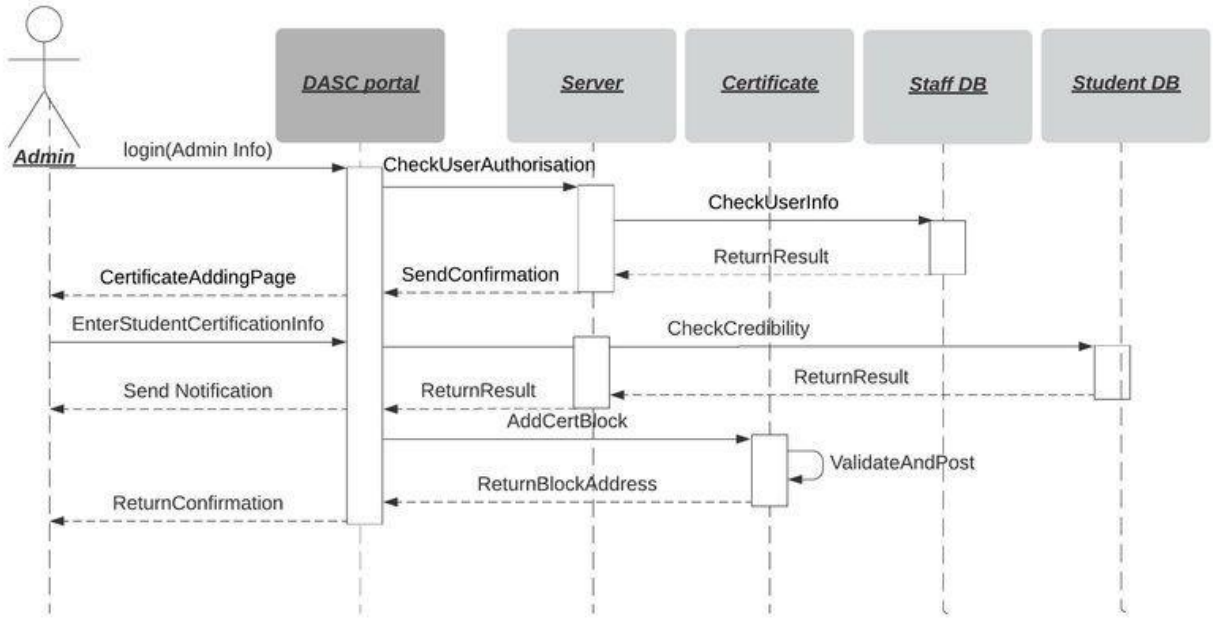
Class Diagram:

Represents the structure of the system, including classes like Certificate, NFT, User, and Blockchain.



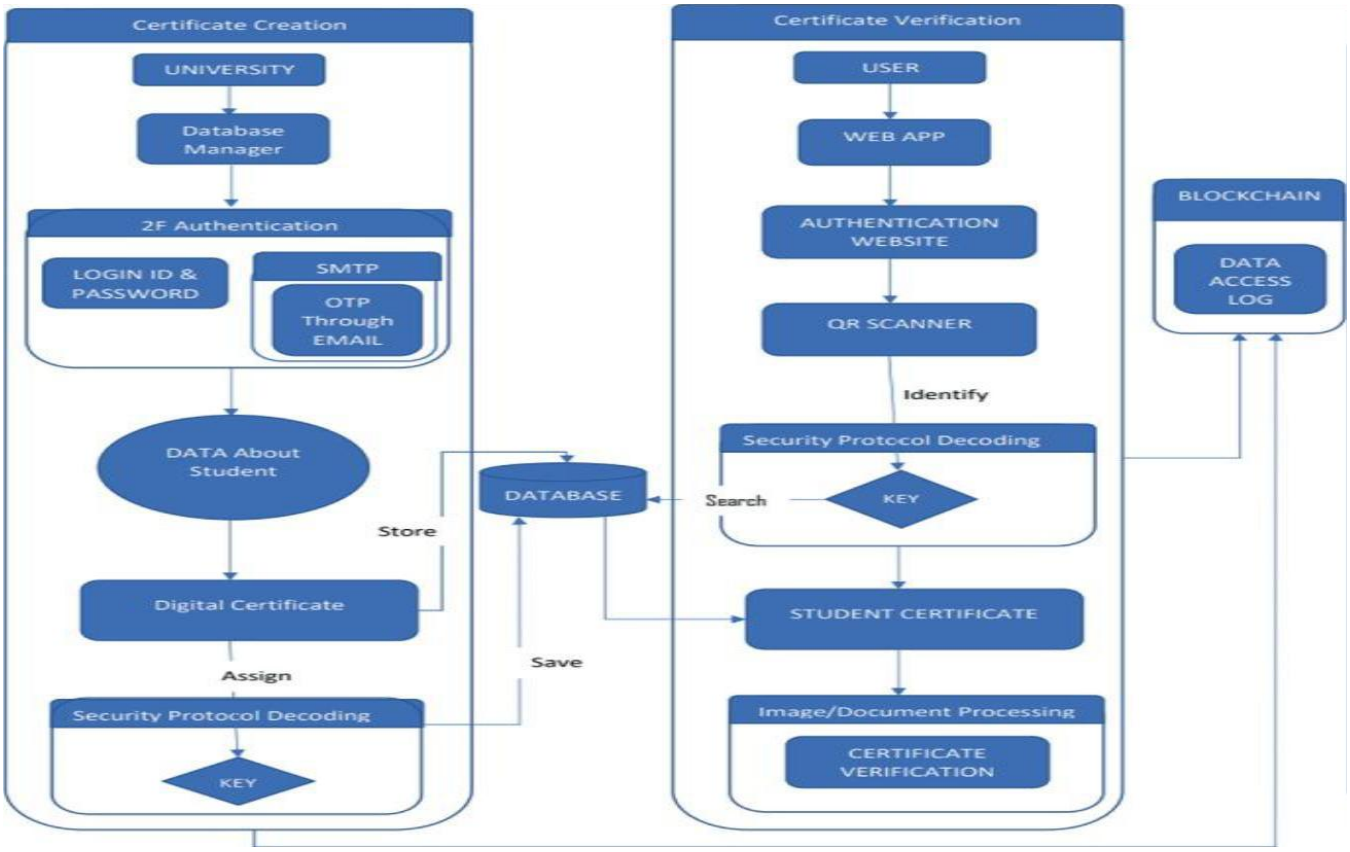
Sequence Diagram:

Illustrates the step-by-step flow from certificate issuance to verification.



Activity Diagram:

Shows the workflow of user interactions and system processes.



Conclusion:

Project Summary:

- A secure, blockchain-based system for academic credential verification.
- Combines blockchain immutability, AI-powered fraud detection, and NFT-based ownership tracking.
- Ensures trust, transparency, and efficiency in credential verification.

Impact:

- Reduces fraud, saves time, and enhances trust in academic credentials.

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for the **Blockchain-Based Certificate Verification System**. This system aims to provide a secure, immutable, and publicly verifiable method for issuing and verifying digital certificates using blockchain technology. The system is designed to reduce fraud and counterfeiting in digital certificates, ensuring quick and anti-counterfeit verification.

1.2 Document Conventions

This document follows the IEEE SRS template. Requirements are numbered sequentially and are categorized by system features. High-priority requirements are marked with an asterisk (*). The document uses the following typographical conventions:

- **Bold text** for section headings and key terms.
- *Italic text* for emphasis or special notes.
- Monospace font for code snippets or technical terms.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- **Developers:** Responsible for implementing the system based on the requirements outlined in this document.
- **Project Managers:** Oversee the development and deployment of the system.
- **Educational Institutions:** Will use the system to issue and verify academic certificates.
- **Employers:** Will use the system to verify the academic credentials of job applicants.
- **Students:** Will use the system to share their academic certificates with potential employers.

It is recommended that readers start with the **Introduction** and **Overall Description** sections to gain a high-level understanding of the system before diving into the detailed system features and requirements.

1.4 Product Scope

The **Blockchain-Based Certificate Verification System** is a cloud-based solution that leverages blockchain technology to issue and verify digital certificates. The system is designed to replace traditional manual verification processes, which are time-consuming

and prone to errors. The system will be used by educational institutions to issue digital certificates, by employers to verify the authenticity of these certificates, and by students to share their academic credentials with potential employers.
The system will provide the following key benefits:

- **Immutability:** Certificates stored on the blockchain cannot be altered or tampered with, ensuring their authenticity.
- **Transparency:** All transactions on the blockchain are publicly verifiable, providing a high level of trust in the system.
- **Efficiency:** The system automates the verification process, reducing the time and effort required to verify certificates.
- **Security:** The use of cryptographic hashing and smart contracts ensures that the system is secure and resistant to fraud.

1.5 References

- IEEE SRS Template
- Blockchain Technology in Education: A Review by D. Ji et al.
- Smart Contracts and Blockchain by Curmi et al.
- Blockchain-Based Cybersecurity by Dasgupta et al.
- SHA-256 Hashing Algorithm Documentation
- Solidity Programming Language Documentation

2. Overall Description

2.1 Product Perspective

The **Blockchain-Based Certificate Verification System** is a new, self-contained product designed to replace traditional manual certificate verification processes. It integrates with existing educational systems and provides a secure, decentralized method for issuing and verifying digital certificates. The system is built on blockchain technology, which ensures that all certificates are immutable and publicly verifiable. The system is designed to be used by educational institutions, employers, and students, and it will be accessible via a web-based interface.

2.2 Product Functions

The system will provide the following key functions:

- **Certificate Issuance:** Educational institutions can issue digital certificates to students upon course completion. The certificates will be stored on the blockchain, ensuring their immutability and authenticity.
- **Certificate Verification:** Employers and institutions can verify the authenticity of certificates using blockchain technology. The system will provide a quick and efficient method for verifying certificates, reducing the time and effort required for manual verification.
- **Smart Contract Execution:** The system will use smart contracts to automate the verification process. Smart contracts will be executed on the blockchain, ensuring that the verification process is secure and tamper-proof.

- **QR Code Generation:** The system will generate QR codes for each certificate, allowing for quick and secure verification. The QR codes will contain encoded information about the certificate, which can be scanned and verified using a smartphone or QR code scanner.

2.3 User Classes and Characteristics

The system will be used by the following user classes:

- **Students:** Users who receive and share their academic certificates. Students will use the system to upload their certificates and share them with potential employers.
- **Educational Institutions:** Users who issue certificates and verify their authenticity. Educational institutions will use the system to issue digital certificates to students and verify the authenticity of certificates issued by other institutions.
- **Employers:** Users who verify the academic credentials of job applicants. Employers will use the system to verify the authenticity of certificates submitted by job applicants.

2.4 Operating Environment

The system will operate in a cloud-based environment and will be compatible with major operating systems (Windows, macOS, Linux). It will require internet access and will support modern web browsers. The system will be built using the following technologies:

- **Blockchain:** Ethereum or Hyperledger for certificate storage and verification.
- **Smart Contracts:** Written in Solidity for automated verification.
- **Web Development:** Next.js for the front-end interface.
- **Database:** Firebase for real-time data management.
- **Backend:** Node.js SDK for server-side logic.

2.5 Design and Implementation Constraints

The system must adhere to the following design and implementation constraints:

- **Blockchain Technology:** The system must use blockchain technology for immutability and transparency.
- **Smart Contracts:** Smart contracts must be written in Solidity and deployed on a blockchain network.
- **QR Code Generation:** The system must support QR code generation for certificate verification.
- **Cryptographic Hashing:** The system must use SHA-256 hashing to ensure data integrity.
- **Scalability:** The system must be scalable to support a large number of users and certificates.

2.6 User Documentation

The following user documentation will be provided:

- **User Manual:** A comprehensive guide for using the system, including instructions for issuing and verifying certificates.

- **Online Help:** Context-sensitive help available within the system.
- **Tutorials:** Step-by-step tutorials for common tasks, such as issuing certificates and verifying certificates.
- **API Documentation:** Documentation for developers who wish to integrate the system with other applications.

2.7 Assumptions and Dependencies

The system makes the following assumptions:

- **Accurate Data:** Educational institutions will provide accurate certificate data.
- **Blockchain Network:** The system depends on a blockchain network for secure and immutable record-keeping.
- **Internet Access:** Users will have reliable internet access to use the system.

---

3. External Interface Requirements

3.1 User Interfaces

The system will provide a web-based interface for users to issue and verify certificates. The interface will include the following components:

- **Dashboard:** A dashboard for educational institutions to issue certificates and view issued certificates.
- **Verification Portal:** A portal for employers to verify the authenticity of certificates.
- **Student Portal:** A portal for students to upload and share their certificates with potential employers.

3.2 Hardware Interfaces

The system will interface with standard hardware devices such as scanners and printers for QR code generation and verification. The system will also support the use of smartphones for scanning QR codes.

3.3 Software Interfaces

The system will integrate with the following software components:

- **Blockchain Networks:** Ethereum or Hyperledger for certificate storage and verification.
- **Cloud Storage:** Firebase for real-time data management.
- **Web Browsers:** The system will be compatible with modern web browsers such as Chrome, Firefox, and Safari.

3.4 Communications Interfaces

The system will use HTTPS for secure communication between clients and servers. It will also support RESTful APIs for integration with third-party systems.

4. System Features

4.1 Certificate Issuance and Verification

4.1.1 Description and Priority

This feature allows educational institutions to issue digital certificates and employers to verify them. It is of high priority.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** Student completes a course.
- **Response:** Institution issues a digital certificate and stores it on the blockchain.

4.1.3 Functional Requirements

- **REQ-1:** The system must allow institutions to issue digital certificates.
- **REQ-2:** The system must allow employers to verify certificates using blockchain technology.

4.2 Smart Contract Execution

4.2.1 Description and Priority

This feature automates the verification process using smart contracts. It is of high priority.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Employer requests certificate verification.
- **Response:** Smart contract executes and returns verification results.

4.2.3 Functional Requirements

- **REQ-3:** The system must execute smart contracts for certificate verification.
- **REQ-4:** The system must generate QR codes for verified certificates.

4.3 QR Code Generation and Verification

4.3.1 Description and Priority

This feature generates QR codes for certificates and allows for quick verification. It is of medium priority.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** Employer scans a QR code.
- **Response:** System displays certificate details and verification status.

4.3.3 Functional Requirements

- **REQ-5:** The system must generate QR codes for each certificate.
- **REQ-6:** The system must allow users to scan QR codes for verification.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system must process certificate verification requests within **5 seconds** under normal operating conditions.
- The system must support at least **100 concurrent verification requests** without degradation in performance.
- Database queries for retrieving certificate information from the blockchain must execute within **2 seconds**.

5.2 Safety Requirements

- The system must ensure that sensitive data is encrypted during transmission and storage using **AES-256 encryption**.
- The system must use **cryptographic hashing (e.g., SHA-256)** to ensure the integrity of certificate data.
- The system must implement regular backup procedures to ensure data recovery in the event of a failure or data loss.

5.3 Security Requirements

- The system must implement **multi-factor authentication (MFA)** for all users.
- The system must enforce **role-based access control (RBAC)** to ensure that only authorized users can perform specific actions.
- The system must maintain detailed **audit logs** of all user actions, including certificate issuance, verification, and modifications.
- The system must undergo regular **vulnerability scans** to identify and address potential security weaknesses.

5.4 Software Quality Attributes

- The system must have a **99.9% uptime**, ensuring high availability.
- The system must be **scalable** to support a large number of users and certificates.
- The system must provide an **intuitive and user-friendly interface**, with clear instructions and minimal learning curve for users.
- The system must be designed for **ease of maintenance**, with modular code and comprehensive documentation.

5.5 Business Rules

- Only **authorized educational institutions** can issue certificates.
- Only **authorized employers** can verify certificates.
- Certificate data must be retained for a minimum of **10 years** to comply with legal and regulatory requirements.
- The system must comply with all relevant **data protection regulations**, including GDPR and FERPA.

6. Other Requirements

- The system must comply with international data protection regulations (e.g., GDPR).

Appendix A: Glossary

- **Blockchain:** A decentralized digital ledger used to record transactions.
- **Smart Contract:** Self-executing contracts with the terms of the agreement directly written into code.
- **QR Code:** A machine-readable code consisting of black and white squares, used for storing information.

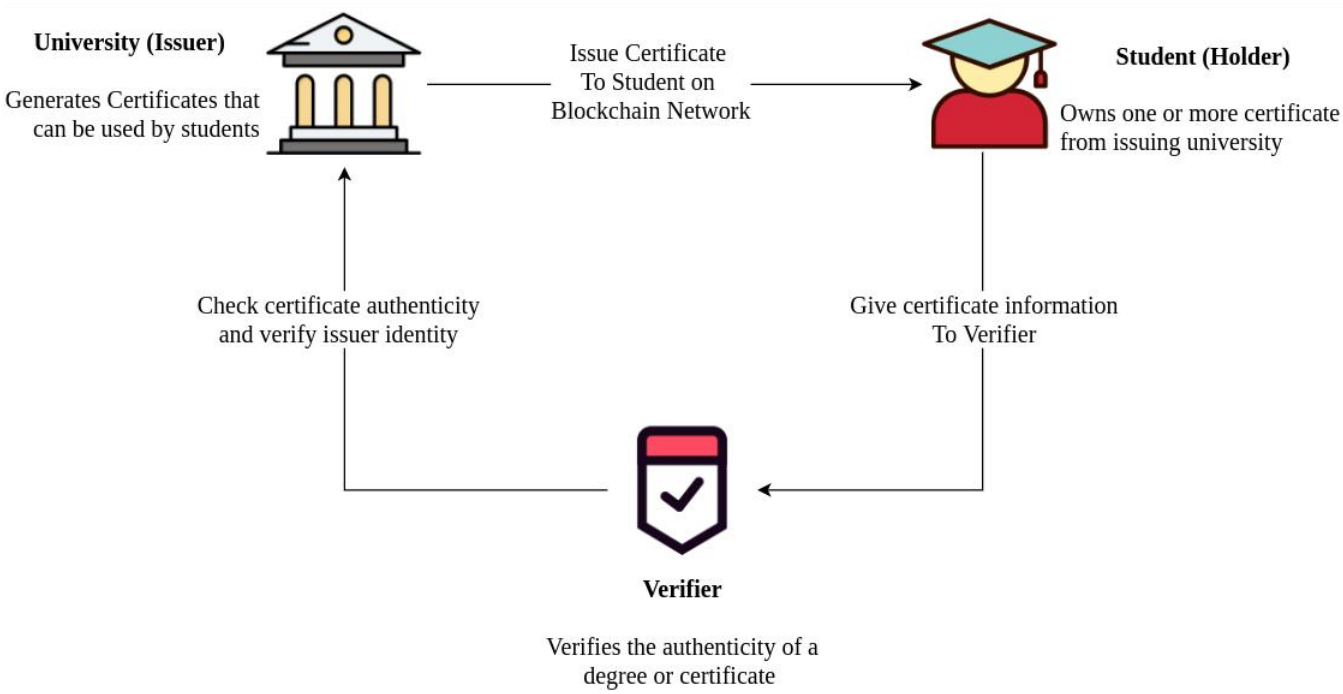
Appendix B: Analysis Models

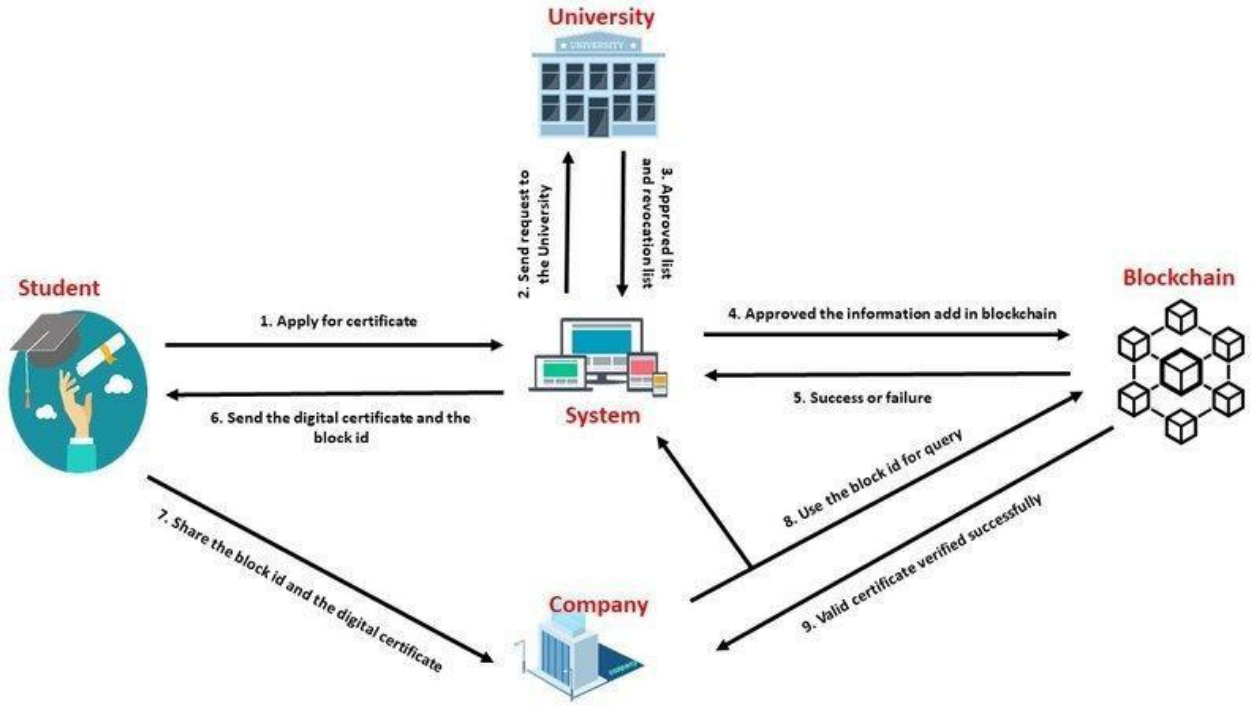
- Data Flow Diagram
- Use Case Diagram

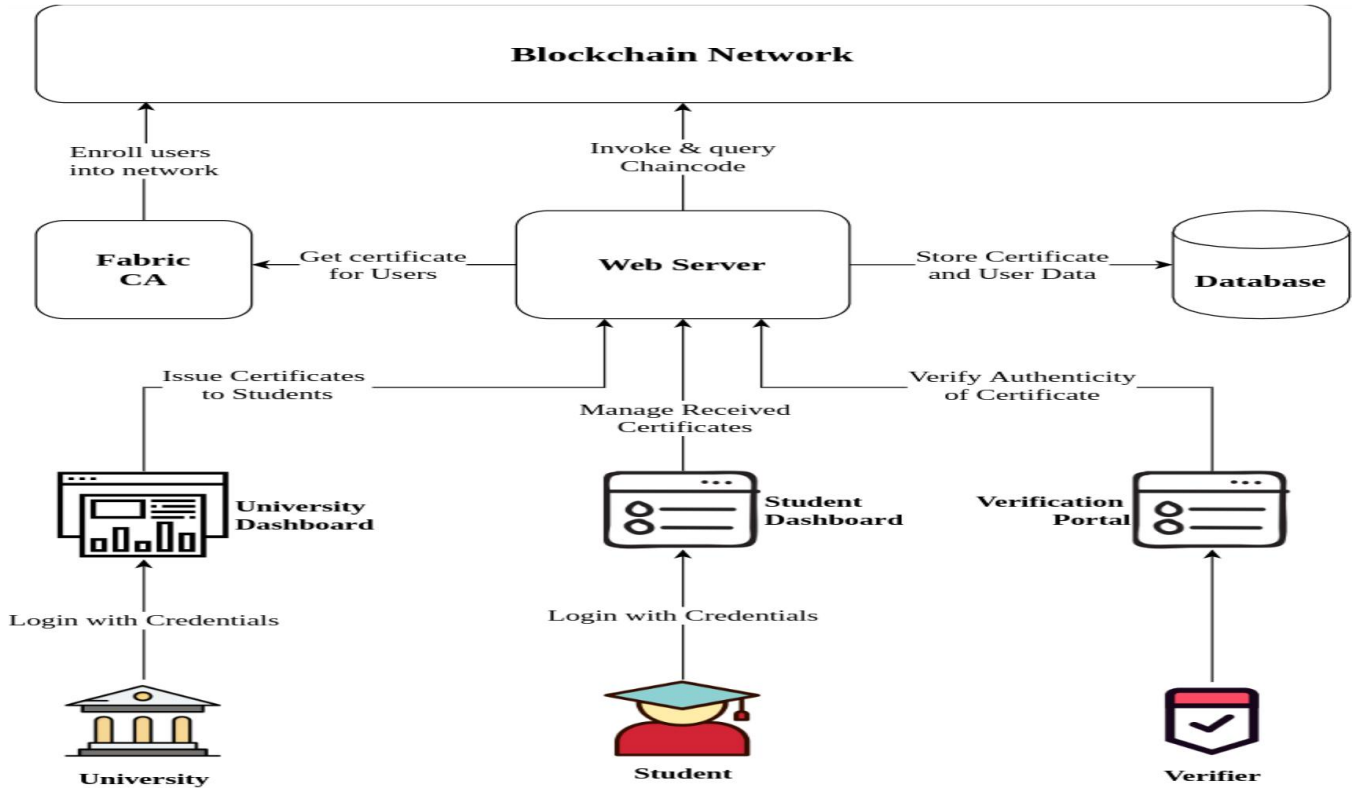
Appendix C: To Be Determined List

- TBD-1: Finalize the blockchain network to be used (Ethereum, Hyperledger, etc.).
- TBD-2: Determine the exact cryptographic hashing algorithm to be used.

Flowchart of the system




University
2. Send request to the University
3. Approved list and revocation list
Student
1. Apply for certificate
6. Send the digital certificate and the block id
System
4. Approved the information add in blockchain
5. Success or failure
Blockchain
8. Use the block id for query
9. Valid certificate verified successfully
7. Share the block id and the digital certificate
Company


Blockchain Network
Enroll users into network
Invoke & query Chaincode
Fabric CA
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Web Server
Store Certificate and User Data
Database
Issue Certificates to Students
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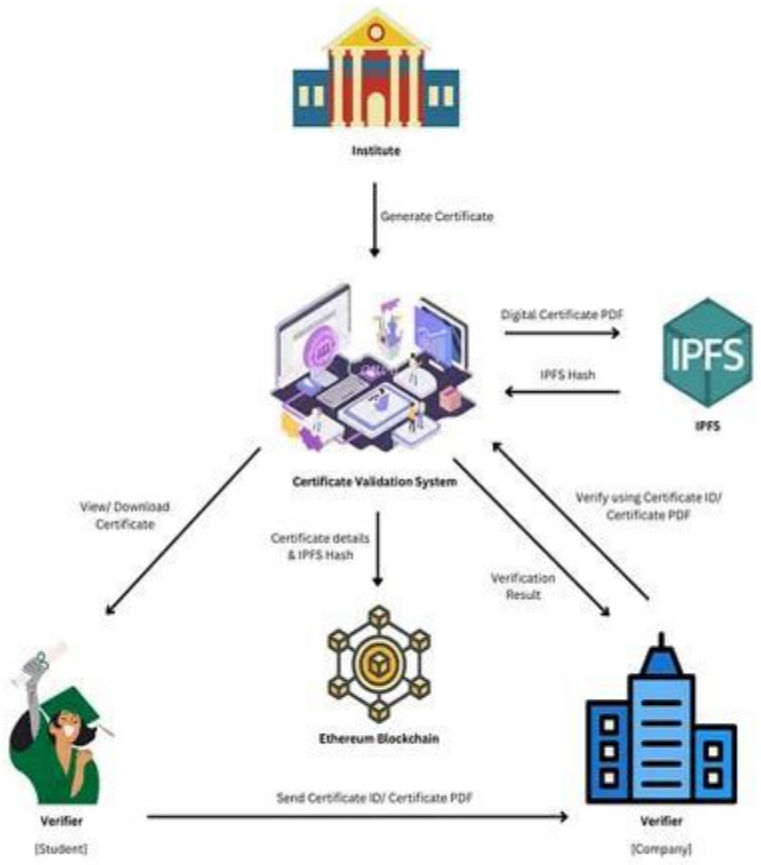
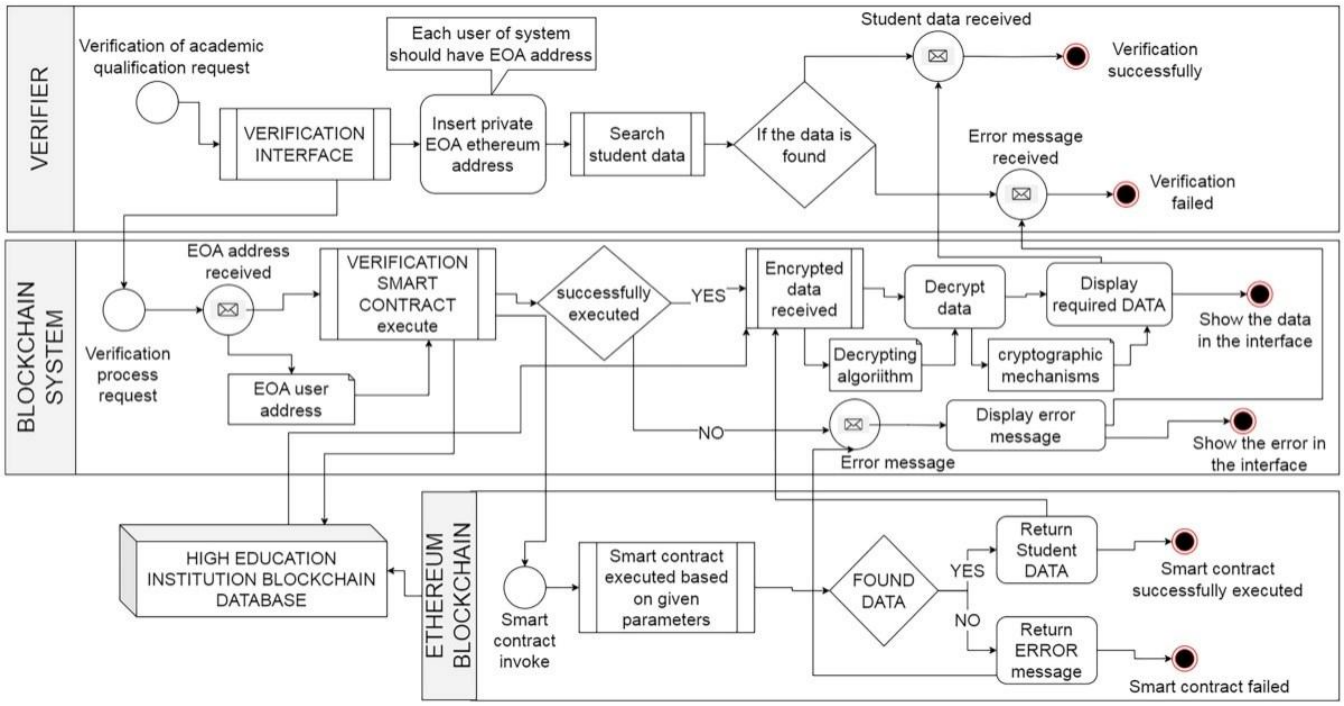


Design - UML Diagrams

Class Diagram:

Represents the structure of the system, including classes like Certificate, NFT, User, and Blockchain.



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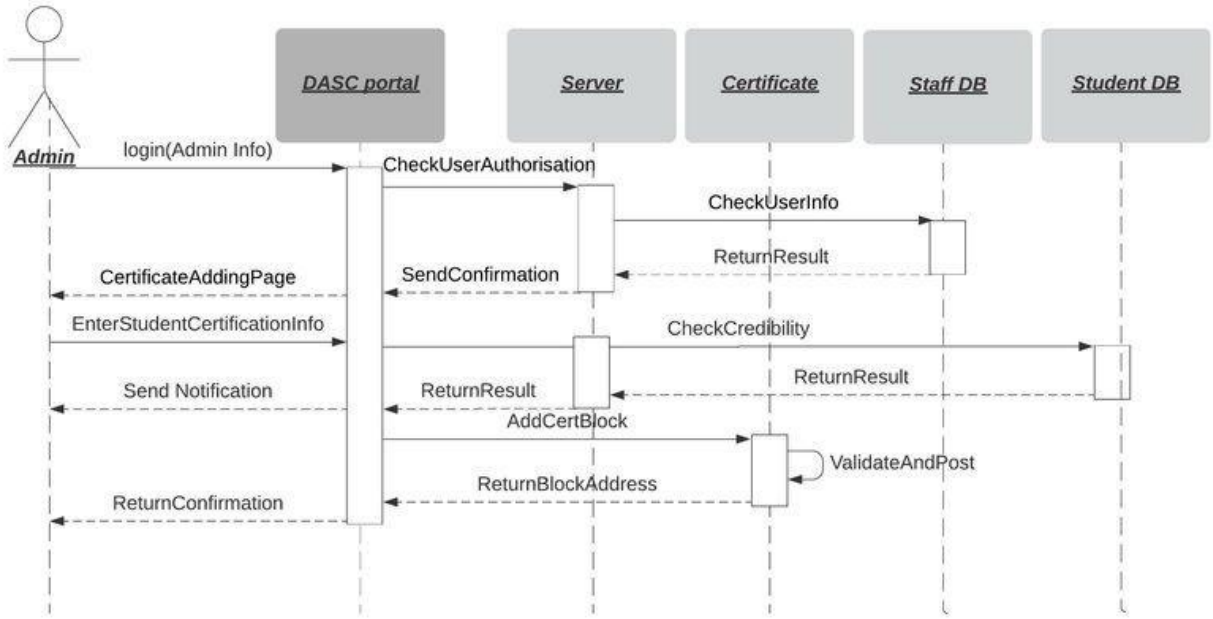
Illustrates the step-by-step flow from certificate issuance to verification.



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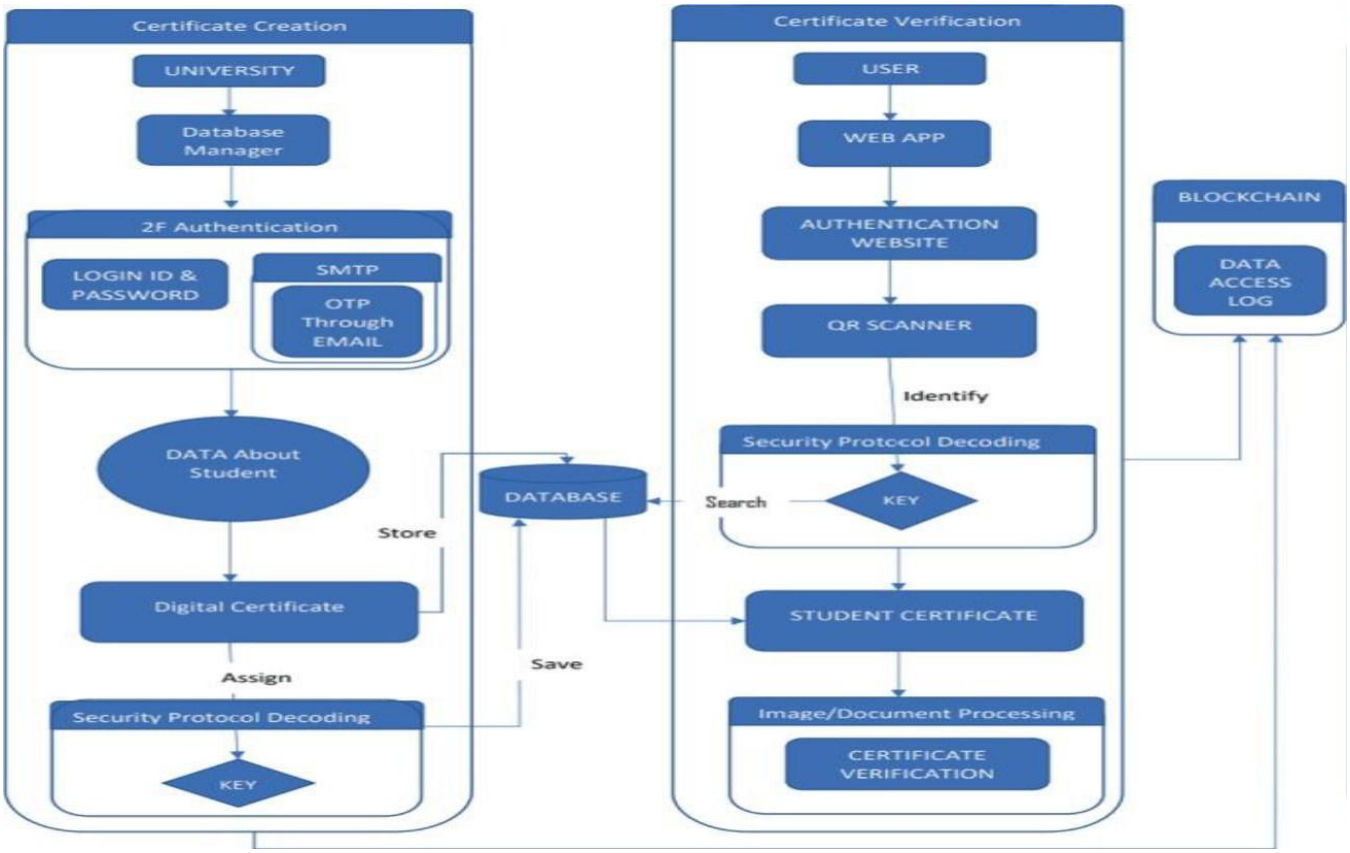

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   2.4 Operating Environment
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   4.3 QR Code Generation and Verification

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1. Introduction

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- **Employers:** Will use the system to verify the academic credentials of job applicants.
- **Students:** Will use the system to share their academic certificates with potential employers.

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1.4 Product Scope

The **Blockchain-Based Certificate Verification System** is a cloud-based solution that leverages blockchain technology to issue and verify academic certificates. The system is designed to replace traditional manual verification processes, which are time-consuming and prone to errors. The system will be used by educational institutions to issue digital certificates, by employers to verify the authenticity of these certificates, and by students to share their academic credentials with potential employers.

The system will provide the following key benefits:

- **Immutability:** Certificates stored on the blockchain cannot be altered or tampered with, ensuring their authenticity.
- **Transparency:** All transactions on the blockchain are publicly verifiable, providing a high level of trust in the system.
- **Efficiency:** The system automates the verification process, reducing the time and effort required to verify certificates.
- **Security:** The use of cryptographic hashing and smart contracts ensures that the system is secure and resistant to fraud.

1.5 References

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- Blockchain-Based Cybersecurity by Dasgupta et al.
- SHA-256 Hashing Algorithm Documentation
- Solidity Programming Language Documentation

2. Overall Description

2.1 Product Perspective

The **Blockchain-Based Certificate Verification System** is a new, self-contained product designed to replace traditional manual certificate verification processes. It integrates with existing educational systems and provides a secure, decentralized method for issuing and verifying academic certificates. The system is built on blockchain technology, which ensures that all certificates are immutable and publicly verifiable. The system is designed to be used by educational institutions, employers, and students, and it will be accessible via a web-based interface.

2.2 Product Functions

The system will provide the following key functions:

- **Certificate Issuance:** Educational institutions can issue digital certificates to students upon course completion. The certificates will be stored on the blockchain, ensuring their immutability and authenticity.
- **Certificate Verification:** Employers and institutions can verify the authenticity of certificates using blockchain technology. The system will provide a quick and

efficient method for verifying certificates, reducing the time and effort required for manual verification.

- **Smart Contract Execution:** The system will use smart contracts to automate the verification process. Smart contracts will be executed on the blockchain, ensuring that the verification process is secure and tamper-proof.
- **QR Code Generation:** The system will generate QR codes for each certificate, allowing for quick and secure verification. The QR codes will contain encoded information about the certificate, which can be scanned and verified using a smartphone or QR code scanner.

2.3 User Classes and Characteristics

The system will be used by the following user classes:

- **Students:** Users who receive and share their academic certificates. Students will use the system to upload their certificates and share them with potential employers.
- **Educational Institutions:** Users who issue certificates and verify their authenticity. Educational institutions will use the system to issue digital certificates to students and verify the authenticity of certificates issued by other institutions.
- **Employers:** Users who verify the academic credentials of job applicants. Employers will use the system to verify the authenticity of certificates submitted by job applicants.

2.4 Operating Environment

The system will operate in a cloud-based environment and will be compatible with major operating systems (Windows, macOS, Linux). It will require internet access and will support modern web browsers. The system will be built using the following technologies:

- **Blockchain:** Ethereum or Hyperledger for certificate storage and verification.
- **Smart Contracts:** Written in Solidity for automated verification.
- **Web Development:** Next.js for the front-end interface.
- **Database:** Firebase for real-time data management.
- **Backend:** Node.js SDK for server-side logic.

2.5 Design and Implementation Constraints

The system must adhere to the following design and implementation constraints:

- **Blockchain Technology:** The system must use blockchain technology for immutability and transparency.
- **Smart Contracts:** Smart contracts must be written in Solidity and deployed on a blockchain network.
- **QR Code Generation:** The system must support QR code generation for certificate verification.
- **Cryptographic Hashing:** The system must use SHA-256 hashing to ensure data integrity.
- **Scalability:** The system must be scalable to support a large number of users and certificates.

2.6 User Documentation

The following user documentation will be provided:

- **User Manual:** A comprehensive guide for using the system, including instructions for issuing and verifying certificates.
- **Online Help:** Context-sensitive help available within the system.
- **Tutorials:** Step-by-step tutorials for common tasks, such as issuing certificates and verifying certificates.
- **API Documentation:** Documentation for developers who wish to integrate the system with other applications.

2.7 Assumptions and Dependencies

The system makes the following assumptions:

- **Accurate Data:** Educational institutions will provide accurate certificate data.
- **Blockchain Network:** The system depends on a blockchain network for secure and immutable record-keeping.
- **Internet Access:** Users will have reliable internet access to use the system.

3. External Interface Requirements

3.1 User Interfaces

The system will provide a web-based interface for users to issue and verify certificates. The interface will include the following components:

- **Dashboard:** A dashboard for educational institutions to issue certificates and view issued certificates.
- **Verification Portal:** A portal for employers to verify the authenticity of certificates.
- **Student Portal:** A portal for students to upload and share their certificates with potential employers.

3.2 Hardware Interfaces

The system will interface with standard hardware devices such as scanners and printers for QR code generation and verification. The system will also support the use of smartphones for scanning QR codes.

3.3 Software Interfaces

The system will integrate with the following software components:

- **Blockchain Networks:** Ethereum or Hyperledger for certificate storage and verification.
- **Cloud Storage:** Firebase for real-time data management.
- **Web Browsers:** The system will be compatible with modern web browsers such as Chrome, Firefox, and Safari.

3.4 Communications Interfaces

The system will use HTTPS for secure communication between clients and servers. It will also support RESTful APIs for integration with third-party systems.

4. System Features

4.1 Certificate Issuance and Verification

4.1.1 Description and Priority

This feature allows educational institutions to issue digital certificates and employers to verify them. It is of high priority.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** Student completes a course.
- **Response:** Institution issues a digital certificate and stores it on the blockchain.

4.1.3 Functional Requirements

- **REQ-1:** The system must allow institutions to issue digital certificates.
- **REQ-2:** The system must allow employers to verify certificates using blockchain technology.

4.2 Smart Contract Execution

4.2.1 Description and Priority

This feature automates the verification process using smart contracts. It is of high priority.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Employer requests certificate verification.
- **Response:** Smart contract executes and returns verification results.

4.2.3 Functional Requirements

- **REQ-3:** The system must execute smart contracts for certificate verification.
- **REQ-4:** The system must generate QR codes for verified certificates.

4.3 QR Code Generation and Verification

4.3.1 Description and Priority

This feature generates QR codes for certificates and allows for quick verification. It is of medium priority.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** Employer scans a QR code.
- **Response:** System displays certificate details and verification status.

4.3.3 Functional Requirements

- **REQ-5:** The system must generate QR codes for each certificate.
- **REQ-6:** The system must allow users to scan QR codes for verification.

---

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system must process certificate verification requests within **5 seconds** under normal operating conditions.
- The system must support at least **100 concurrent verification requests** without degradation in performance.
- Database queries for retrieving certificate information from the blockchain must execute within **2 seconds**.

5.2 Safety Requirements

- The system must ensure that sensitive data is encrypted during transmission and storage using **AES-256 encryption**.
- The system must use **cryptographic hashing (e.g., SHA-256)** to ensure the integrity of certificate data.
- The system must implement regular backup procedures to ensure data recovery in the event of a failure or data loss.

5.3 Security Requirements

- The system must implement **multi-factor authentication (MFA)** for all users.
- The system must enforce **role-based access control (RBAC)** to ensure that only authorized users can perform specific actions.
- The system must maintain detailed **audit logs** of all user actions, including certificate issuance, verification, and modifications.
- The system must undergo regular **vulnerability scans** to identify and address potential security weaknesses.

5.4 Software Quality Attributes

- The system must have a **99.9% uptime**, ensuring high availability.
- The system must be **scalable** to support a large number of users and certificates.
- The system must provide an **intuitive and user-friendly interface**, with clear instructions and minimal learning curve for users.
- The system must be designed for **ease of maintenance**, with modular code and comprehensive documentation.

5.5 Business Rules

- Only **authorized educational institutions** can issue certificates.
- Only **authorized employers** can verify certificates.
- Certificate data must be retained for a minimum of **10 years** to comply with legal and regulatory requirements.

- The system must comply with all relevant **data protection regulations**, including GDPR and FERPA.

6. Other Requirements

- The system must comply with international data protection regulations (e.g., GDPR).

Appendix A: Glossary

- **Blockchain:** A decentralized digital ledger used to record transactions.
- **Smart Contract:** Self-executing contracts with the terms of the agreement directly written into code.
- **QR Code:** A machine-readable code consisting of black and white squares, used for storing information.

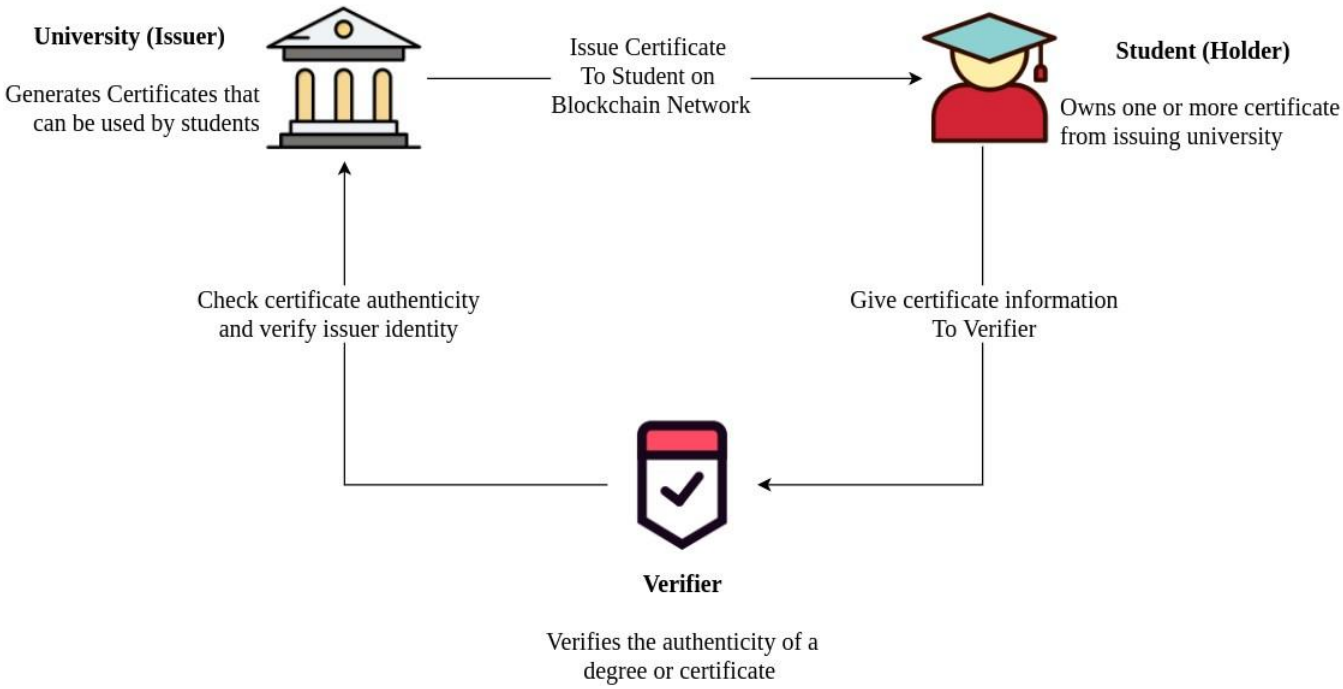
Appendix B: Analysis Models

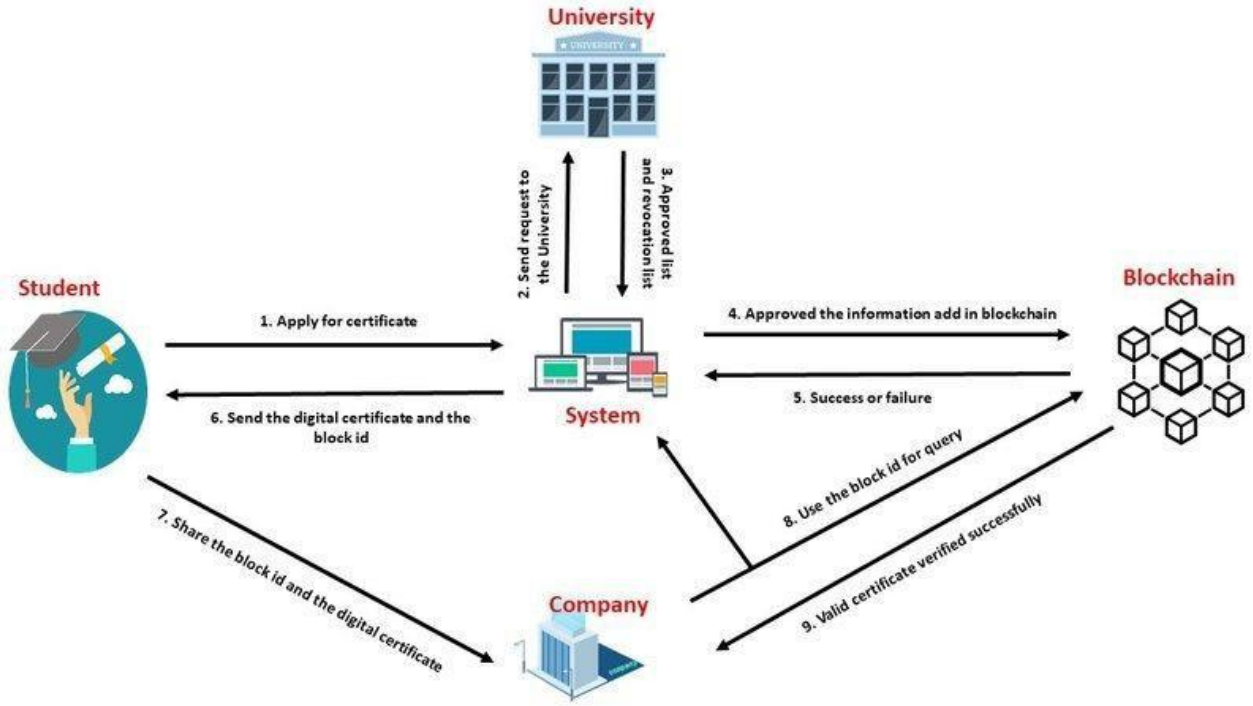
- Data Flow Diagram
- Use Case Diagram

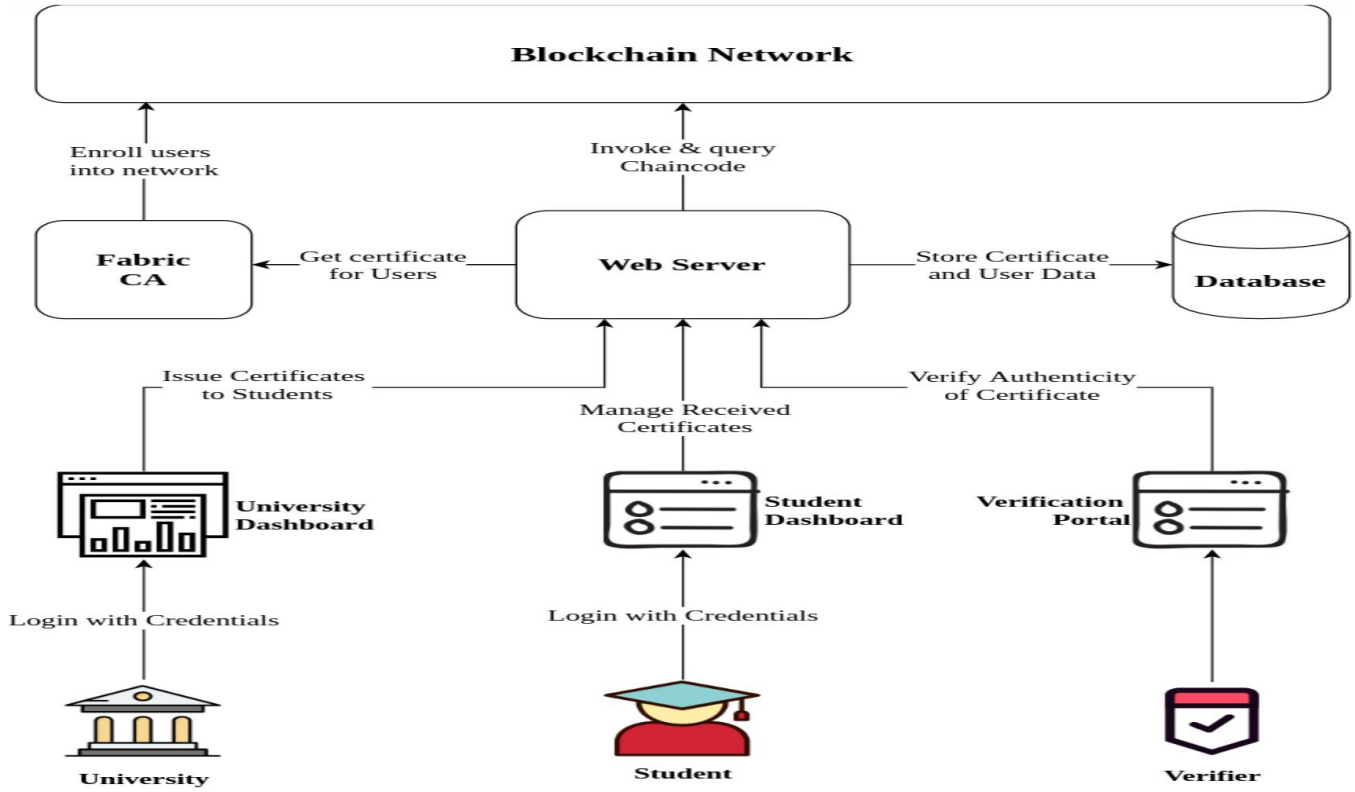
Appendix C: To Be Determined List

- TBD-1: Finalize the blockchain network to be used (Ethereum, Hyperledger, etc.).
- TBD-2: Determine the exact cryptographic hashing algorithm to be used.

Flowchart of the system

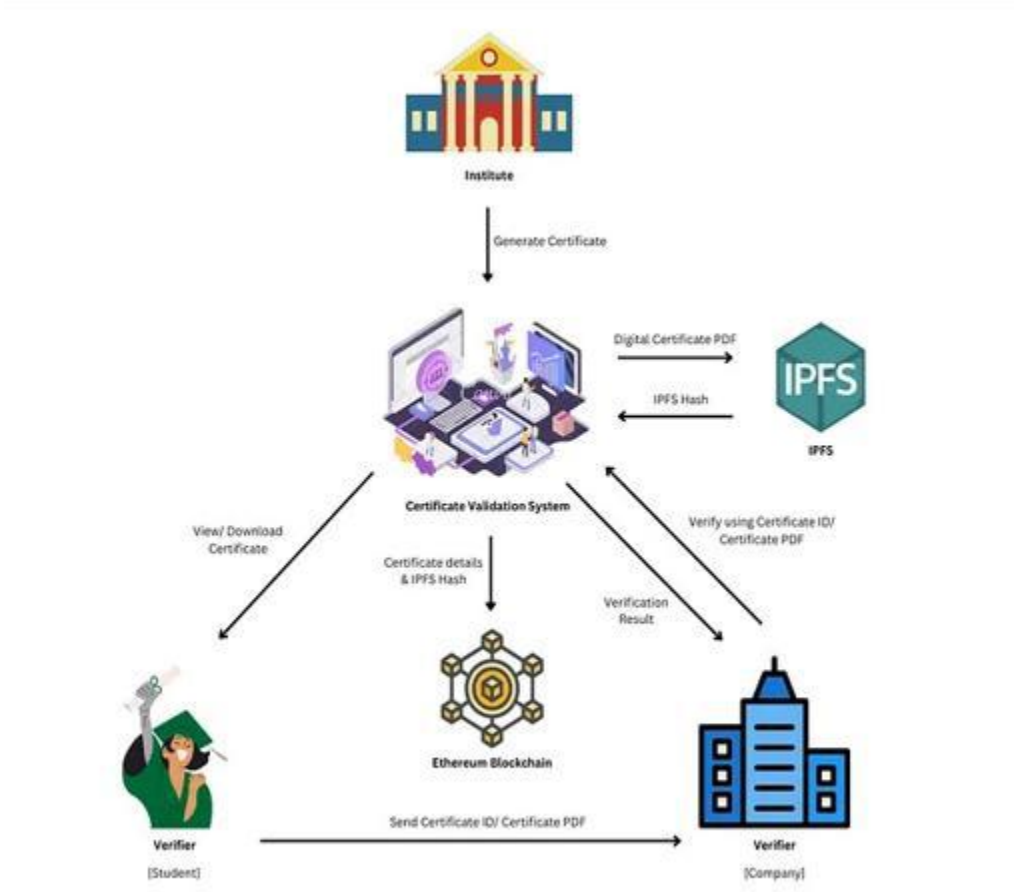



University
2. Send request to the University
3. Approved list and revocation list
Student
1. Apply for certificate
6. Send the digital certificate and the block id
System
4. Approved the information add in blockchain
5. Success or failure
Blockchain
8. Use the block id for query
9. Valid certificate verified successfully
7. Share the block id and the digital certificate
Company


Blockchain Network
Enroll users into network
Invoke & query Chaincode
Fabric CA
Get certificate for Users
Web Server
Store Certificate and User Data
Database
Issue Certificates to Students
Manage Received Certificates
Verify Authenticity of Certificate
University Dashboard
Student Dashboard
Verification Portal
Login with Credentials
Login with Credentials
University
Student
Verifier

Fields	Description
Publisher	IEEE
Date of publish	2023
Authors	Yazan Abu Hammoudeh, Mohammad Qatawneh, Orieb AbuAlghanam, Mohammed A Almaiah.
Pros	- Decentralization, transparency, immutability and security of blockchain techonology. - Reduces fraud and counterfeiting in digital certificates.
Cons	- Challenges in transitioning from traditional systems to blockchain-based systems. - Integration with existing systems can be complex.
Methodology	- Survey of blockchain-based solutions for digital certificate validation. - Analysis of blockchain features and their application in certificate validation.
Accuracy	- 95% (High accuracy in identifying blockchain's potential to reduce fraud and improve certificate validation). - Based on up-to-date research studies.

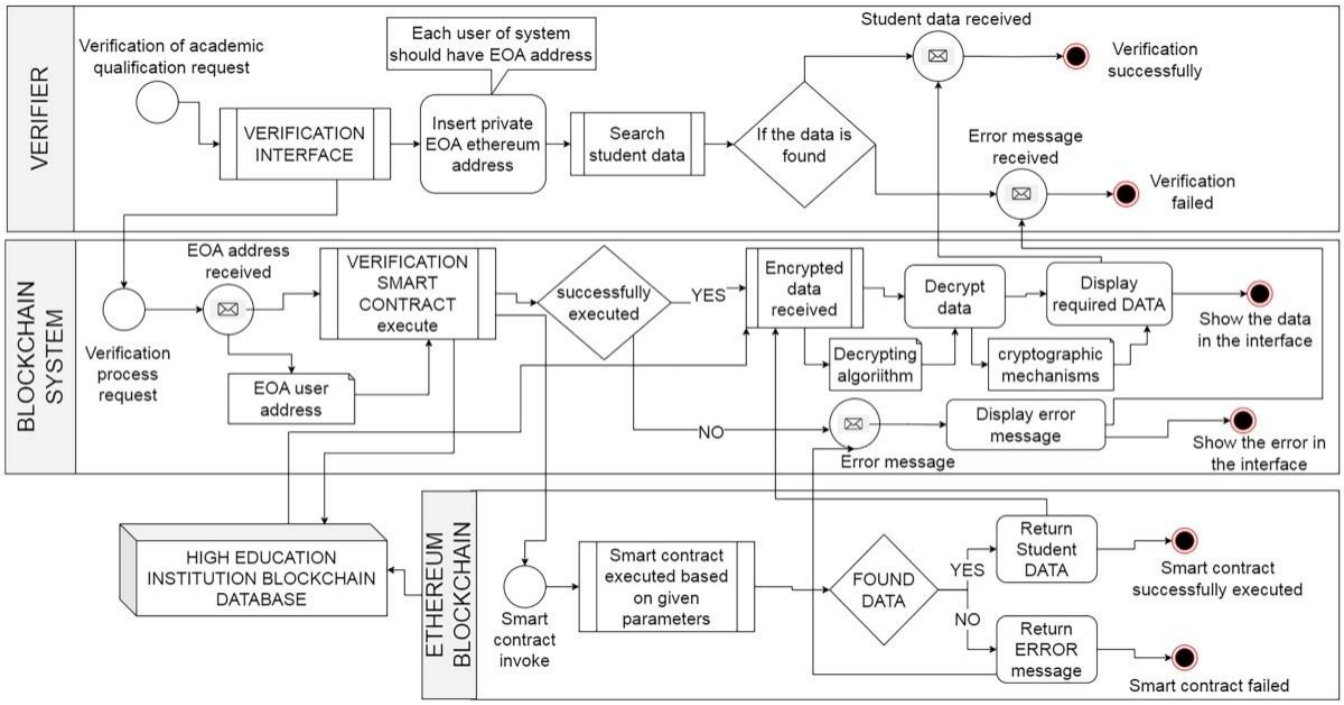
Use Case Diagrams



Design - UML Diagrams

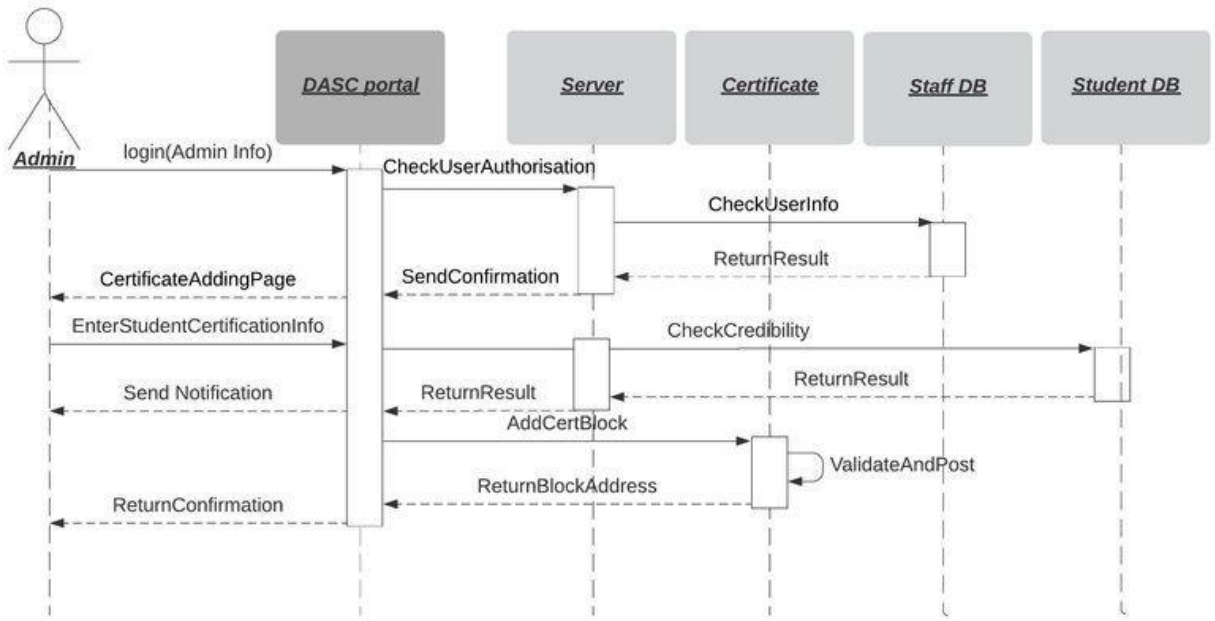
Class Diagram:

Represents the structure of the system, including classes like Certificate, NFT, User, and Blockchain.



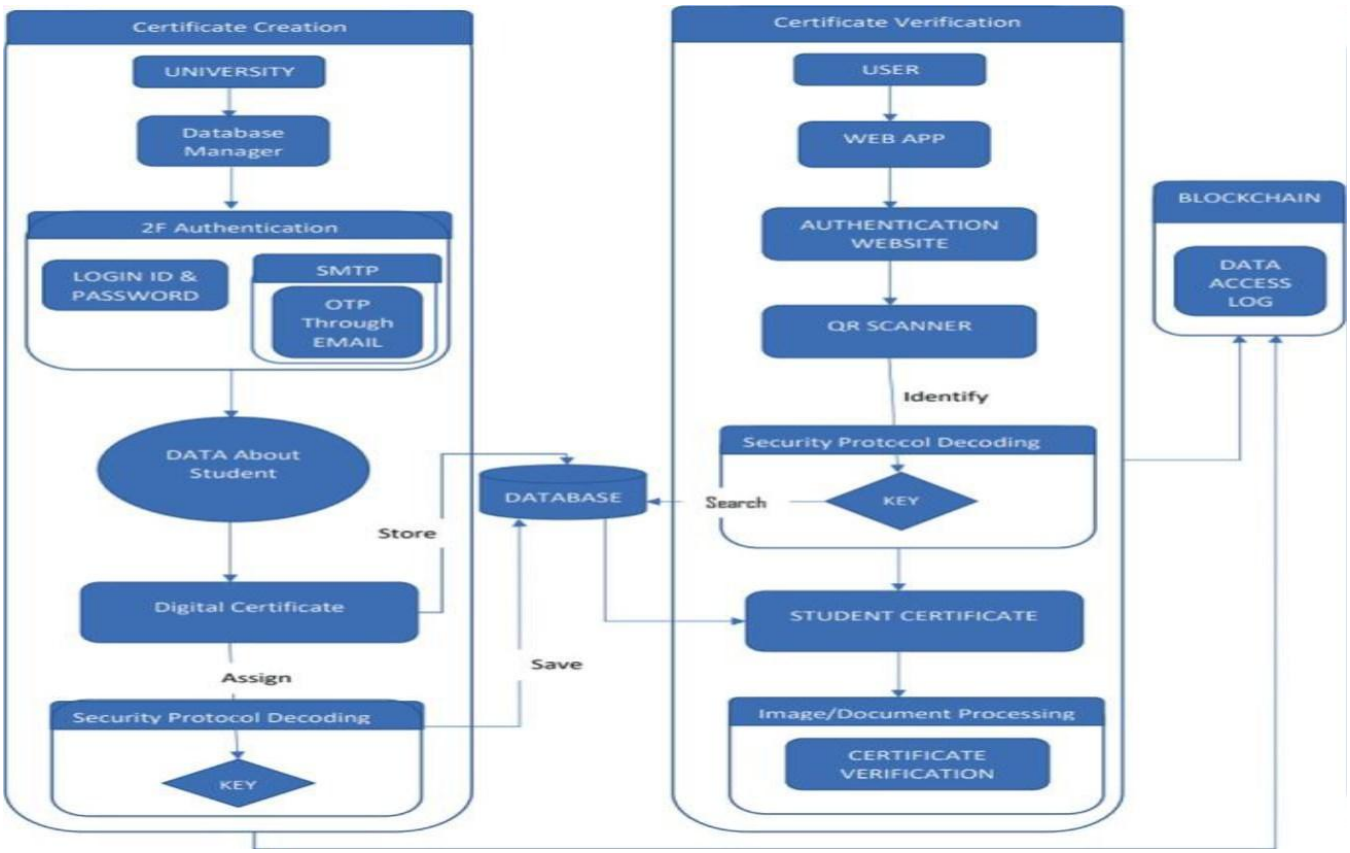
Sequence Diagram:

Illustrates the step-by-step flow from certificate issuance to verification.



Activity Diagram:

Shows the workflow of user interactions and system processes.



Conclusion:

Project Summary:

- A secure, blockchain-based system for academic credential verification.
- Combines blockchain immutability, AI-powered fraud detection, and NFT-based ownership tracking.
- Ensures trust, transparency, and efficiency in credential verification.

Impact:

- Reduces fraud, saves time, and enhances trust in academic credentials.